LAB MANUAL

(CS-314) Computer Networks

By

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Experiment Name: IP address and Sub-netting.

Aim: Study of Network IP

- Introduction of IP Address.
- Classification of IP Address.
- Subnetting.
- Super Netting.
- Installation Cisco Packet Tracer.

Apperatus(Software): N/A Description:

Following is required to be study under this practical

Introduction of IP address:

"A unique numerical label assigned to a device connected to a network that uses the Internet Protocol for communication."

While IP Address is associated with logical addressing and mac Address is associated with physical addressing. IPv4 is 32 bit while Mac Address is 48 bit. Ip address is divided into two types.

Based on Version:

- IPv4: A 32 bit address, the most commonly used IP Format
- IPv6: A 128 bit address, designed to replace IPv4 due to address Exhaustion.

Based on Accessibility:

- Public: A globally unique IP assigned by the Provider which is used as global identification for communication on internet.
- Private: A more locally Assigned IP to communicate within network without routing to internet.

Classification of IP Address: As shown in figure we get the idea regarding how the IP Address are classified and when there use cases.

Class	Address Range	Supports
Class A	1.0.0.0 to 126.255.255.254	Supports 16 million hosts on each of 127 networks
Class B	128.1.0.1 to 191.255.255.254	Supports 65,000 hosts on each of 16,000 networks.
Class C	192.0.1.1 to 223.255.254.254	Supports 254 hosts on each of 2 million networks.
Class D	224.0.0.0 to 239.255.255.255	Reserved for multicast groups.
Class E	240.0.0.0 to 254.255.255.254	Reserved.

Table 1

Note: as u might've noticed that 127.255.255.254 is missing it is because it is used for network loop back.

Subnetting:

The Process of Dividing a large IP Network into smaller, More Manageable sub- networks to improve efficiency, security and performance.

Class	Address Range	Default Mask	Subnet/CIDR Notation
Class A	1.0.0.0 – 126.255.255.255	255.0.0.0	/8
Class B	128.0.0.0 – 191.255.255.255	255.255.0.0	/16
Class C	192.0.0.0 – 223.255.255.255	255.255.255.0	/24

Table 2

Some Formulas:

Formula to calculate Sub-nets:

2^n

Formula to calculate Usable Hosts per subnet :

(2^h)-2

Example Regarding Subneting:

For `192.168.1.0/26` (subnet mask `255.255.255.192`):

1. Subnets: `2^2 = 4` (borrowed 2 bits from host portion).
2. Hosts per subnet: `(2^6)-2 = 62`.

Super netting:

The process to combine multiple smaller sub-nets into a larger network by modifying sub-net mask.

Installation of Cisco Packet Tracer:

1. Open Browser and go to https://www.netacad.com/articles/news/download-cisco- packet-tracer then press download button.

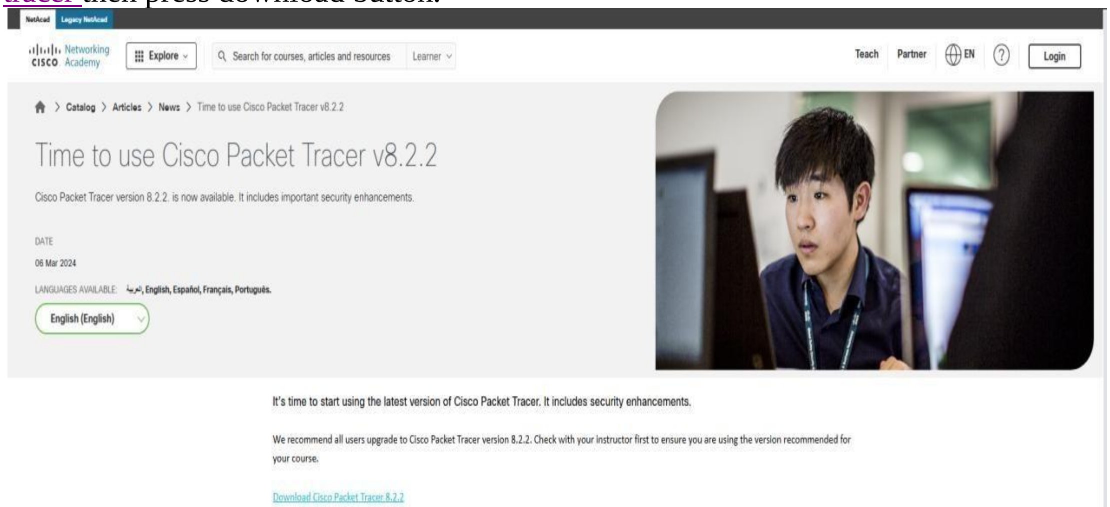


Figure 1

2. After that you will be redirected on the login page where you can use your university Email to register yourself.

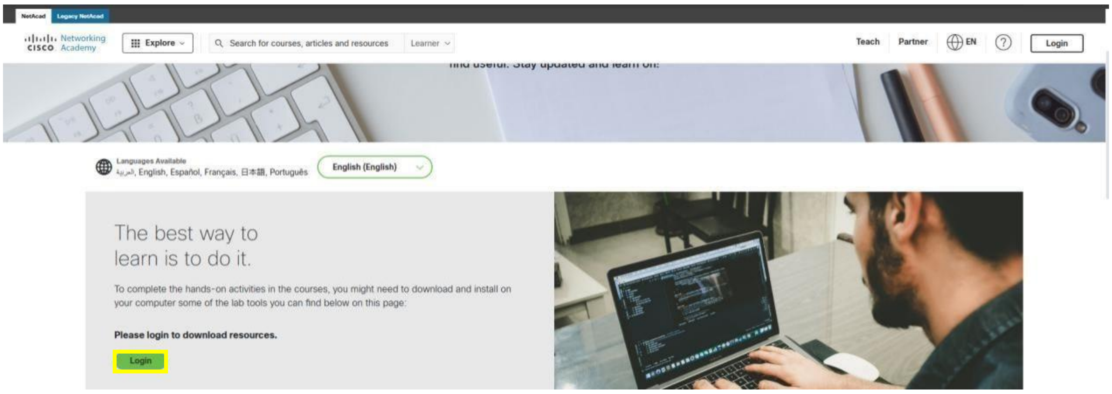


Figure 2

3. Once you register yourself you will be able to download the file.

4. After downloading the file just open the .exe file and follow the installation wizard.

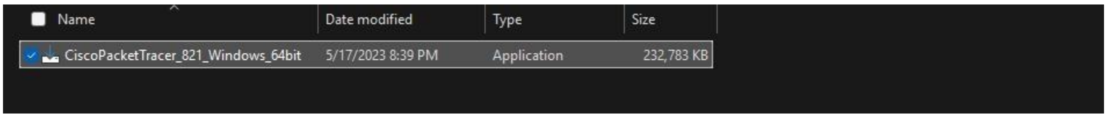


Figure 3

5. It will prompt you to Accept the Digital License Agreement, after accepting it just hit the install button.

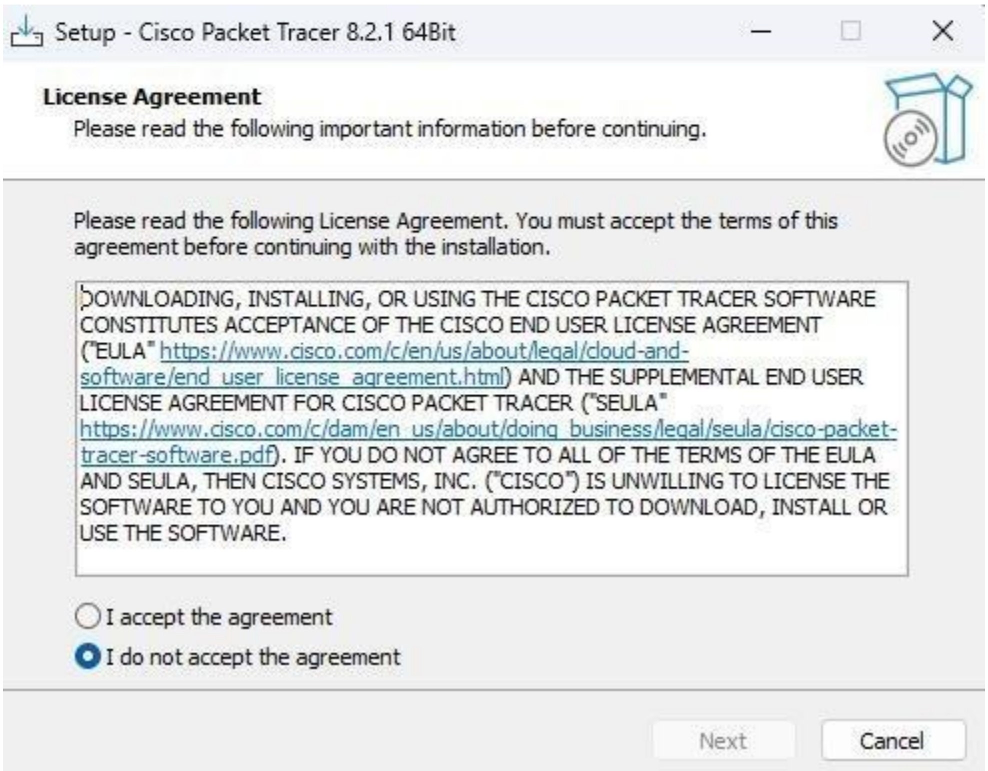


Figure 4

6. Once Installed, Click "Finish" to Complete.

7. After That Launch and login the account to start free trial (or Just Disconnect the network to so it doesn't ask you to login).

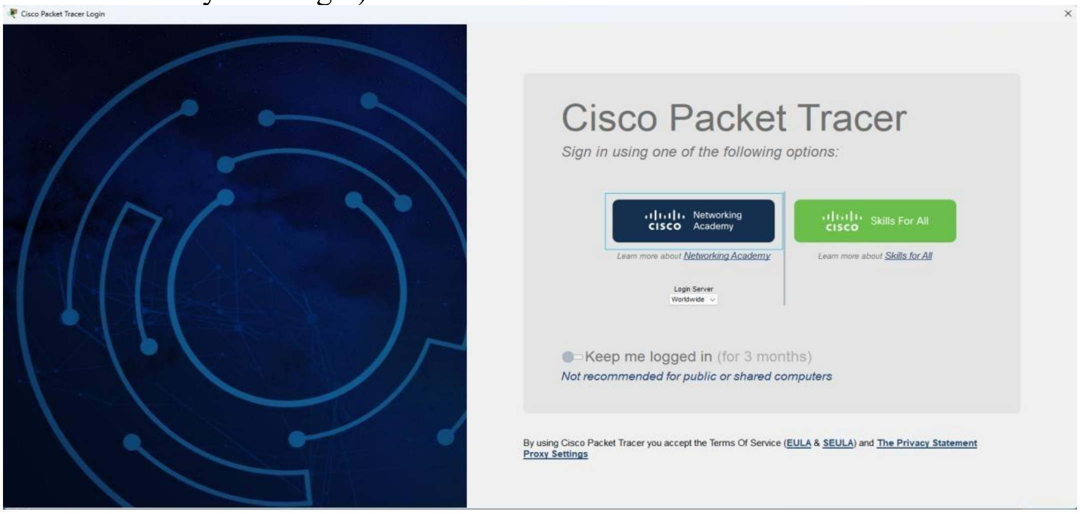


Figure 5

Lab2

Experiment Name: Introduction to Packet Tracer.

Aim: Study of Basic network command and network configuration commands, and establishing a LAN of 2 Computers.

Apparatus (software): Cisco Packet Tracer and Command Prompt.

Description:

Packet Tracer: Packet Tracer is a cross-platform visual simulation tool designed by Cisco Systems that allows users to create network topologies and imitate modern computer networks. The software allows users to simulate the configuration of Cisco routers. Packet Tracer is a powerful network simulator that can be utilized in training for CCNA.

- Learn the basic operations of Packet Tracer: - File commands, visualization and configuration of networking devices.
- Simulate the interactions of data traveling through the network.
- Learn to visualize the network in logical and physical modes.
- Reinforce your understanding with extensive hands-on networking and activities.

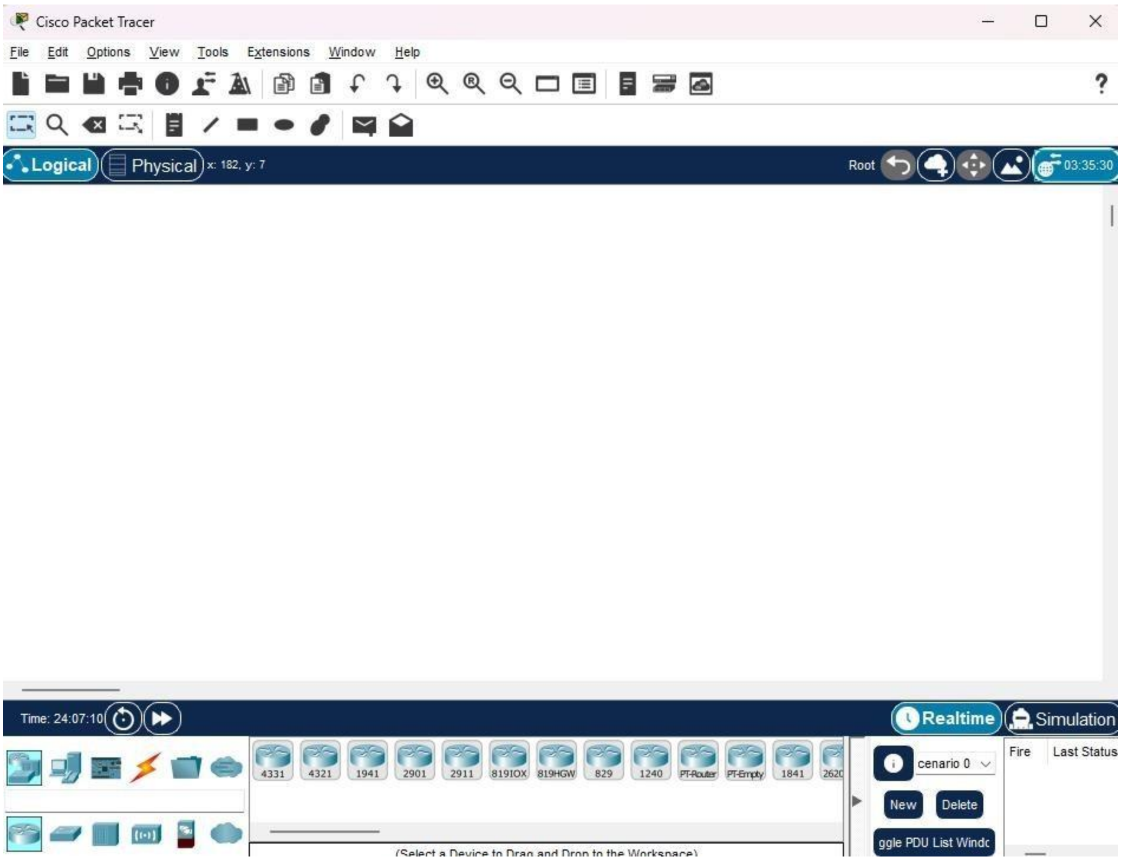


Figure 6

This is the opening window of packet tracer.

We are different modules and panels available in the packet tracer. Some important modules, which are important to understand for the working in Packet Tracer.

We can add many devices and make connection.

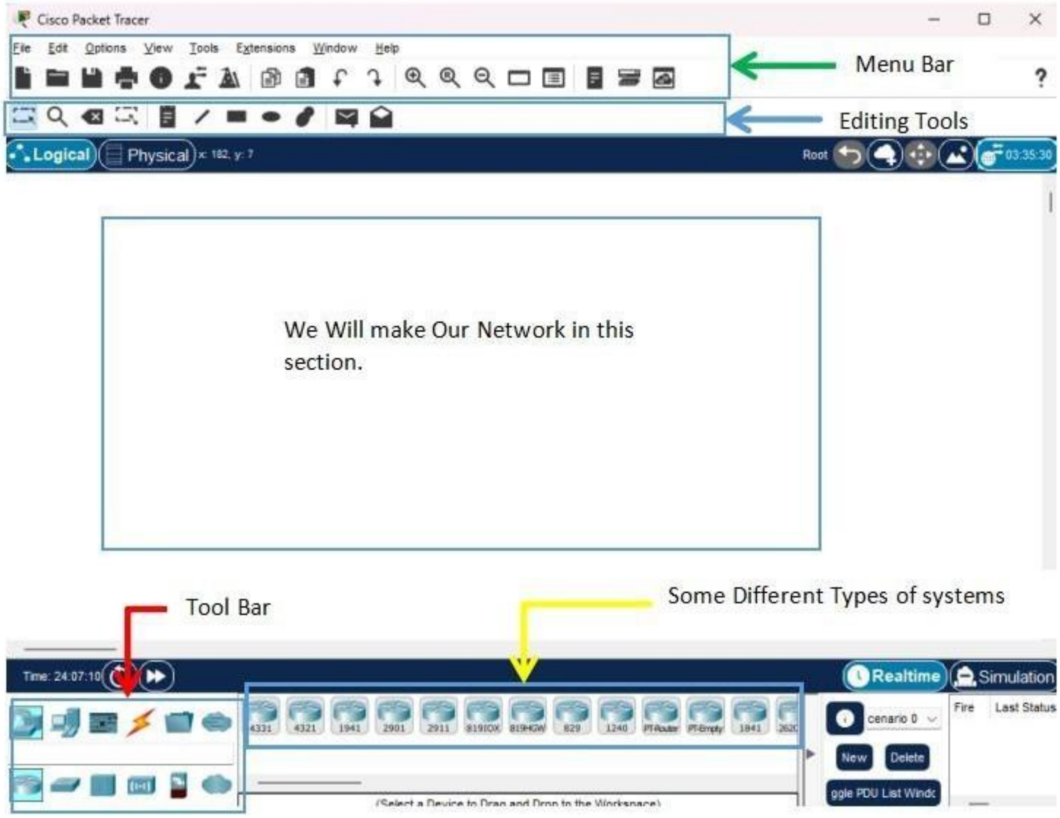


Figure 7

Connection of 2 Systems using copper crossover cable to make a 2 System LAN connection.

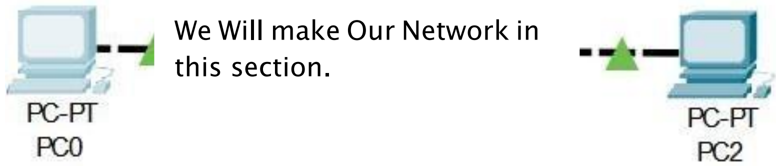


Figure 8

Each System have unique IP address as shown in the following figure.

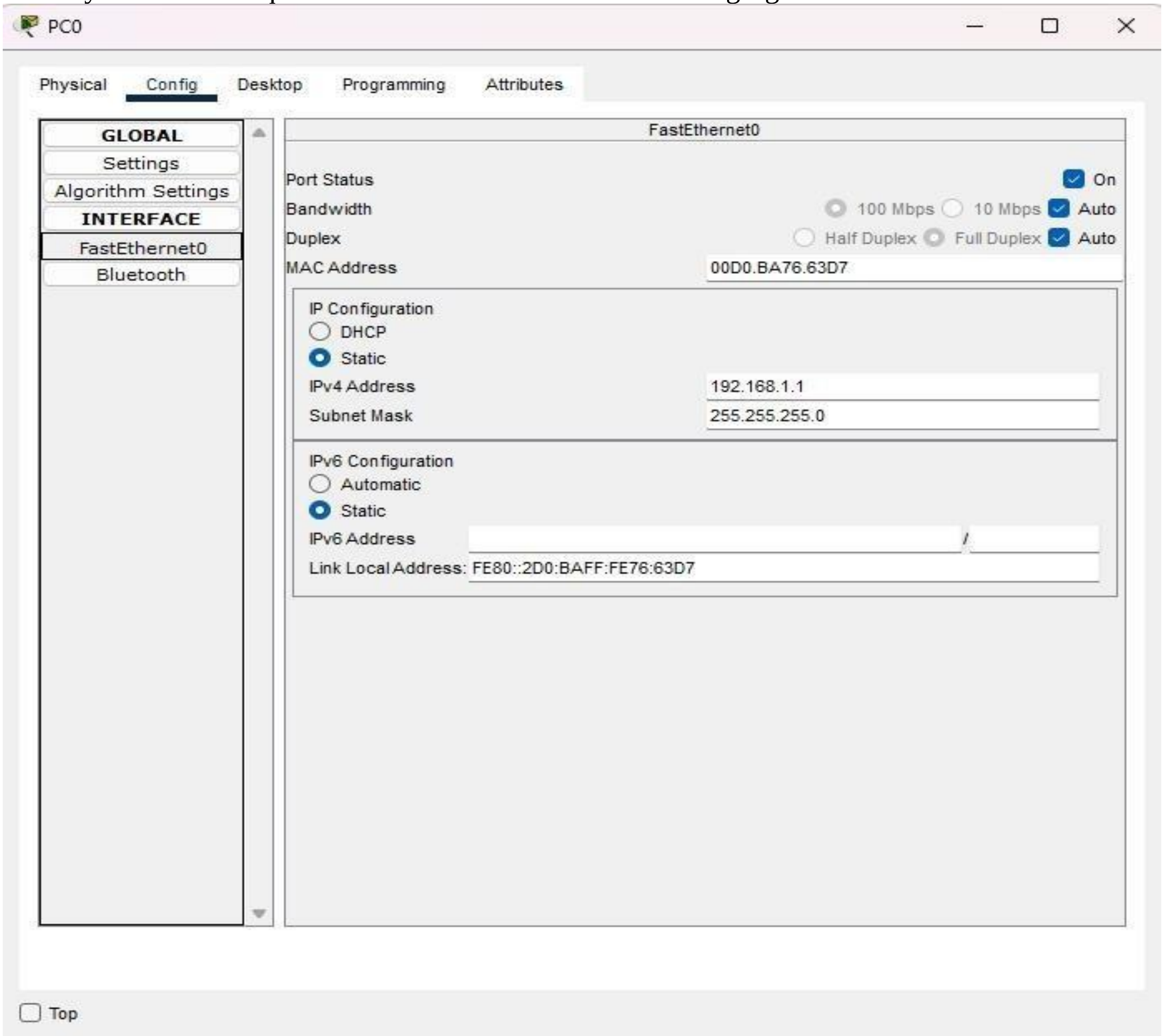


Figure 9

The Message is being shared Successfully.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)

Figure 10

Ping and Tracert Command

- Double Click on the System(PC0 in our case), It will open a dialogue box then navigate to Desktop section.

Figure 11

- Click on the Command prompt section and open it.

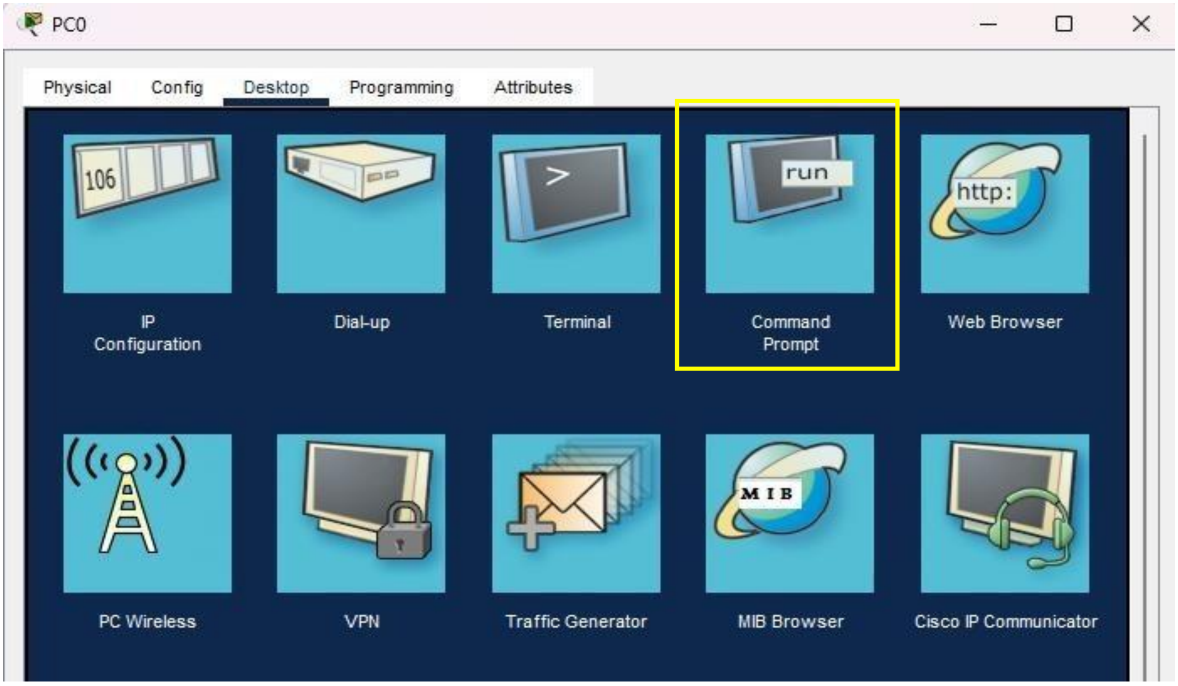


Figure 12

Figure 13

- Using Ping Command to Send Packets to PC1

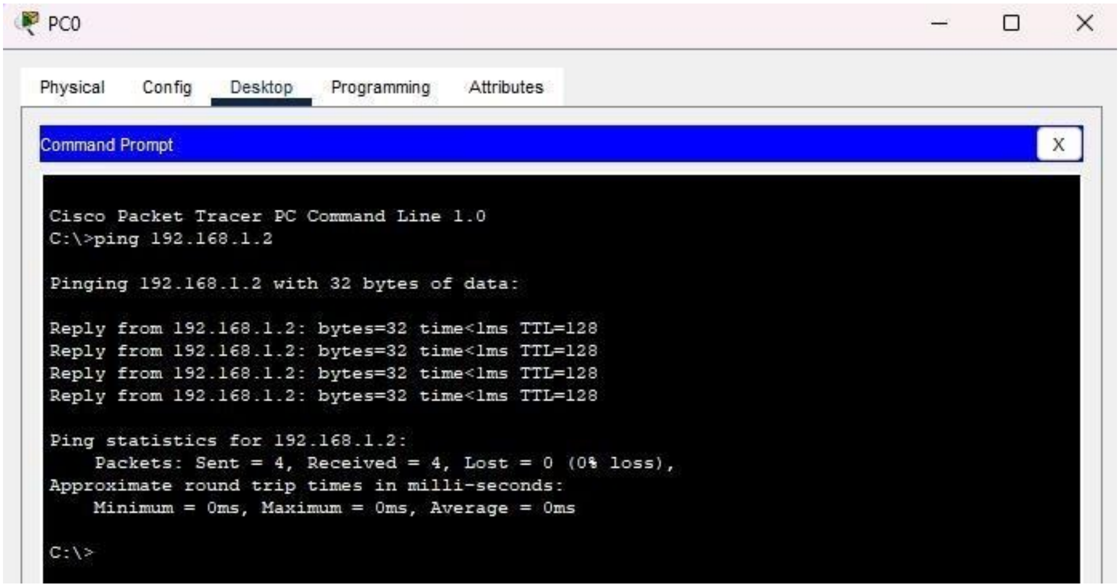


Figure 14

- Using Tracert to Check Route of PC0

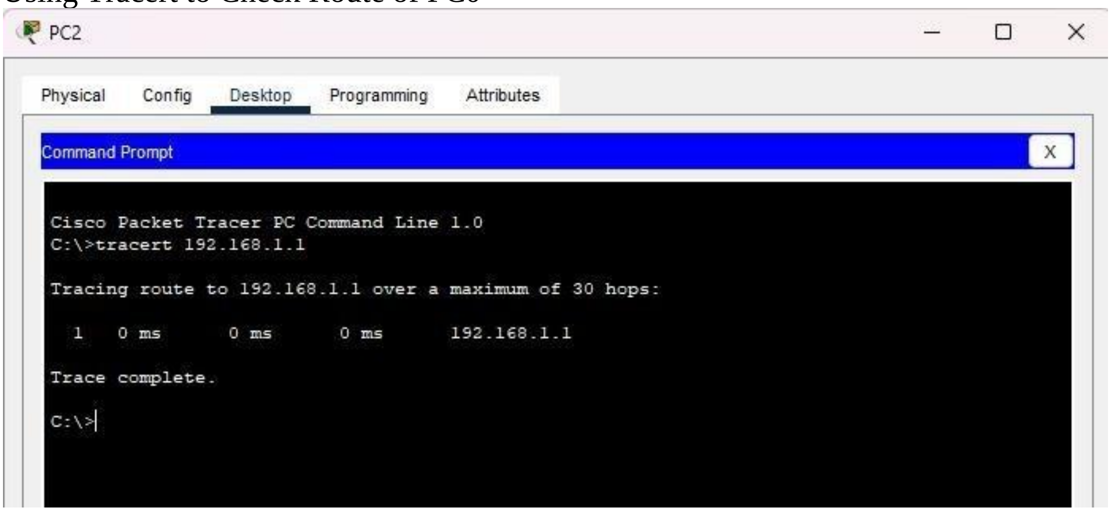


Figure 15

Lab Assignment 1:

Define Hub, Switch,Router and Servers.

Hub:

A hub is one of the most basic and earliest devices used in networking. Fundamentally, a hub is a networking device that connects multiple Ethernet devices, making them act as a single network segment. It operates at the physical layer of the OSI model, meaning it does not manage any of the traffic passing through it. A hub receives incoming packets, often referred to as frames, and broadcasts them to all other ports, regardless of the intended recipient. This means every device connected to the hub receives the same data; however, only the intended recipient processes it further, while others discard it.

This broadcasting method can lead to a significant number of data collisions, especially in networks with many devices, which can degrade overall network performance. Moreover, since all data is sent to every connected device, security can be a concern, as data is not selectively targeted to its destination. Hubs are generally considered legacy technology in modern networks. They are inexpensive and commonly used in small, uncomplicated network environments where network traffic and security are not significant concerns.

Figure 16

Types of Hubs

Following Given are the types of hubs

- **Passive:** Connects multiple devices without any signal amplification
- **Active:** Amplifies and regenerates signal to extend the range of a network
- **Intelligent:** Management features monitoring like monitoring and traffic control

Features:

It has following Features.

- Broadcasting to All Ports
- Not Intelligent
- Physical Layer of OSI model
- Simple and Inexpensive
- Multiple Ports But Shared Bandwidth
- Low Security
- Half Duplex

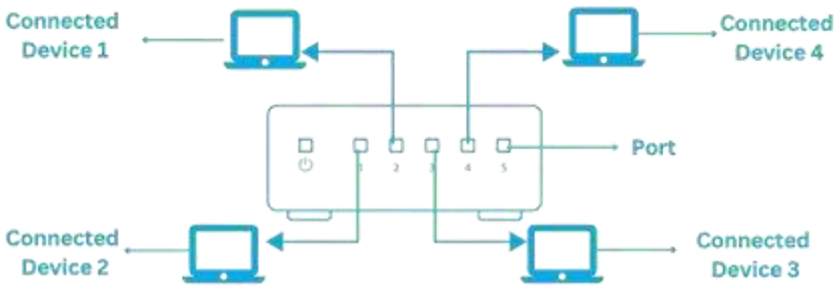


Figure 17

Switch:

A switch operates at a more advanced level compared to a hub. It is a crucial device in most networking environments, functioning at the data link layer (Layer 2) of the OSI model. Unlike a hub, which broadcasts data to all connected devices, a switch intelligently manages data flow by maintaining a MAC address table that tracks the devices connected to each port.

When a data packet arrives at one of the ports, the switch examines the destination MAC address and forwards the packet only to the appropriate port where the recipient device is connected. This prevents unnecessary broadcasting, significantly reducing network congestion and minimizing collisions, thereby improving efficiency and performance.

Switches come with multiple ports, typically 8, 16, 24, or 48, allowing multiple devices to connect and communicate efficiently. By directly sending data to the intended recipient, switches help manage network traffic effectively while also providing a more secure environment compared to hubs.

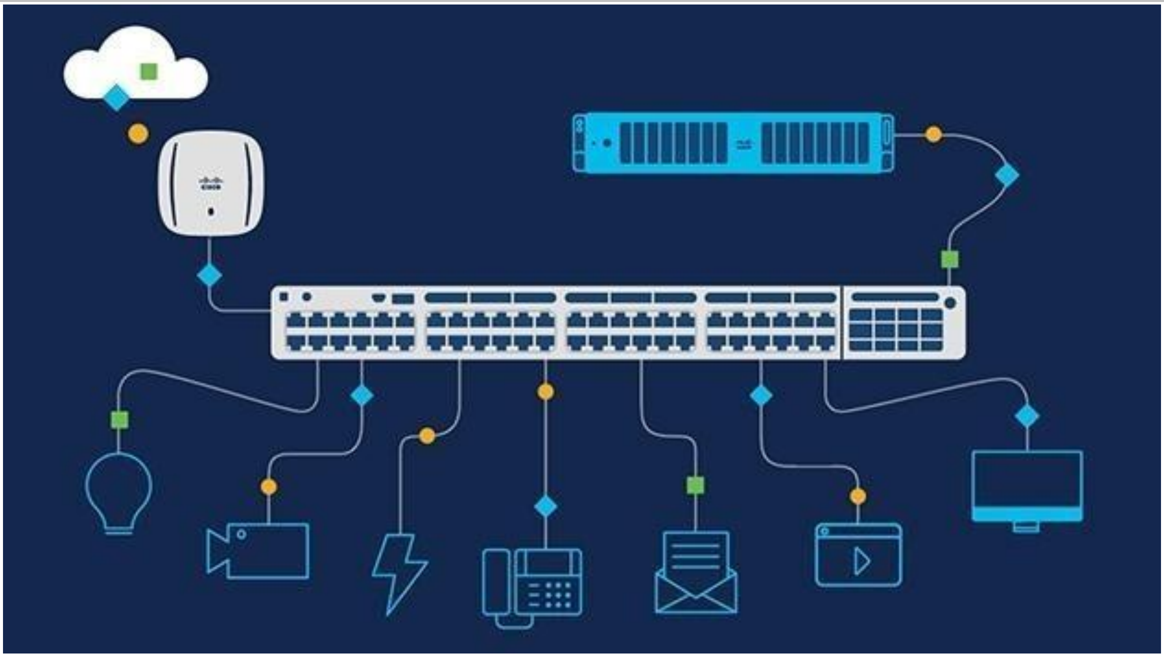


Figure 18

Types of Switches:

Following given are types of switches

- Un-Managed Switches.
- Managed Switches.
- Power Over Ethernet Switches.
- Modular Switches.
- Fixed Switches.
- Lan Switches.
- Stackable Switches.

Features:

It has following Features

- Directs data to appropriate address
- More Intelligent then Hub
- Data Link Layer of OSI Model
- More Expensive then Hubs
- Dedicated Bandwidth for each Port(8-48)
- Single LAN
- Full Duplex

Router:

A router is a sophisticated device that operates at the network layer (Layer 3) of the OSI model and plays a crucial role in the connectivity of modern networks. Routers are designed to connect multiple networks, such as local area networks (LANs) like your home Wi-Fi, wide area networks (WANs), and the internet. By doing so, they facilitate the flow of data between different network segments, making them essential for creating an expansive, interconnected network.

Routers use IP addresses to determine the best path for forwarding data packets. This routing process involves analyzing the destination IP address of each packet and making decisions based on a routing table that contains information about which paths lead to particular network destinations. This capability allows routers to efficiently manage and direct traffic, ensuring that data reaches its intended destination via the most optimal path.

Modern routers often include built-in firewalls and can support connections to virtual private networks (VPNs), providing robust security measures to protect the network from unauthorized access and threats. Additionally, they can connect devices using both wireless (Wi-Fi) and wired (Ethernet) connections, offering flexibility in setting up your network.

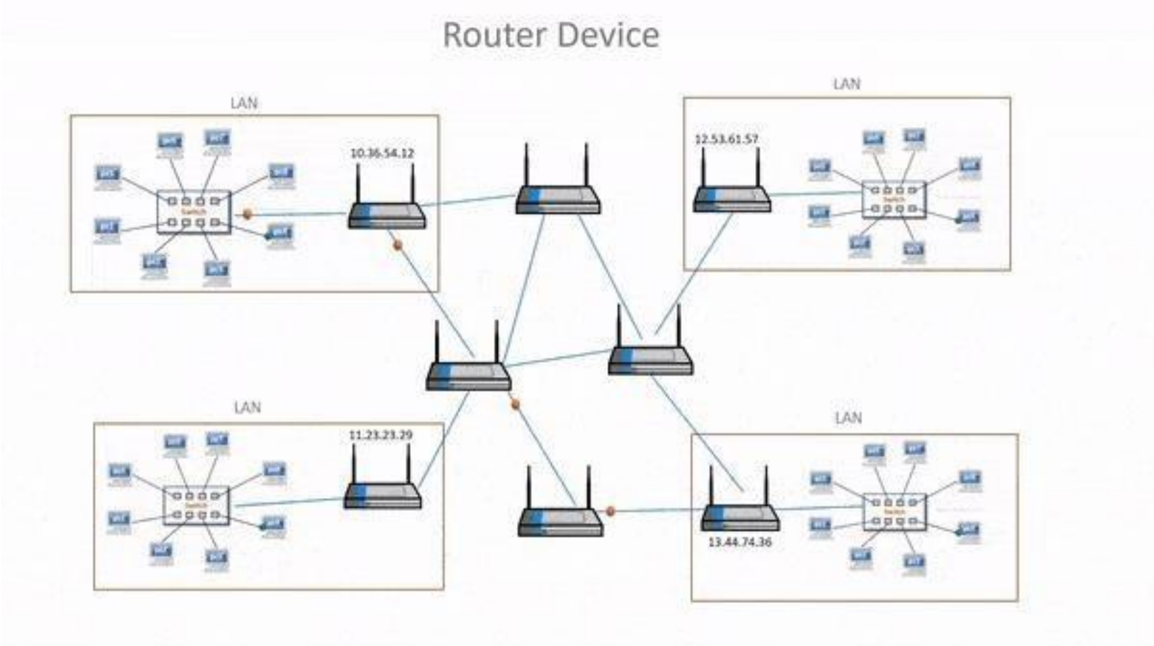


Figure 19

Types of Routers:

Following given are the type of Routers

- Core Routers
- Wireless Routers
- Edge Routers
- VPN Routers
- Branch Routers
- Virtual Routers

Features:

It has following Features

- Routes Data through Multiple networks using IP.

- Most Intelligent
- Operates at network Layer of OSI Model
- Most Expensive
- Fewer Ports including LAN and WAN
- Routes Data Between Networks
- Highly Secure.

Servers:

Servers are large data storage and processing devices that exist either as hardware or as virtual storehouse or as virtual Storehouse located on the internet. Computers or software systems act as servers that connect to the network. A server can be any type of device that shares and saves information. Servers can both store and process information within their own system or request it from another. Servers began as small devices that simply transferred data to a more functional computer then grew in size and ability to perform more complex functions.

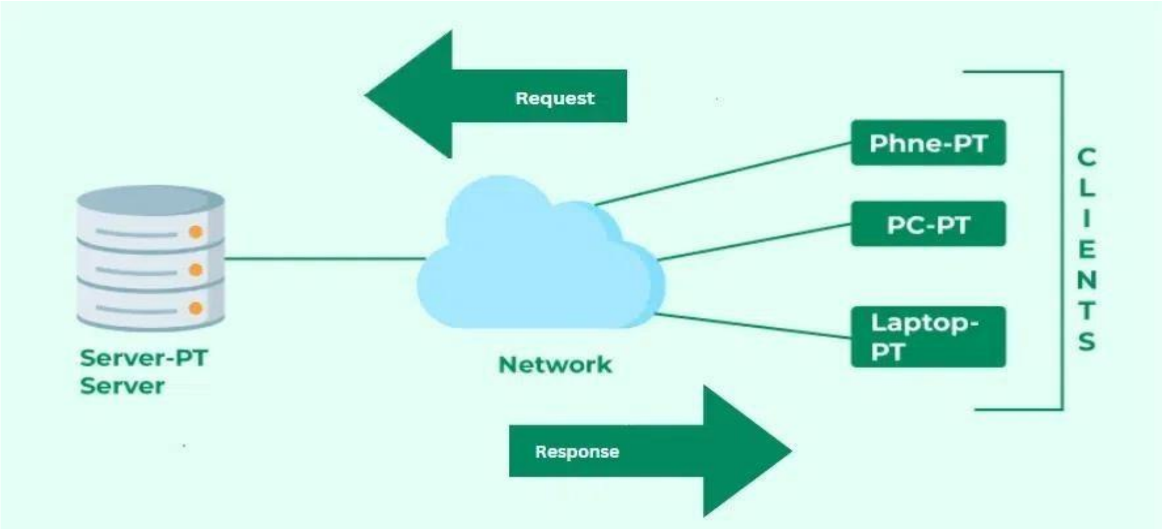


Figure 20

Types of Servers:

Following given are the types of servers

- Web Server
- Proxy Server
- Virtual Machine
- FTP Server
- Application Server
- File Server
- Database Server
- Mail Server
- DNS Server
- Gaming Server

Components of Servers:

Following given are the components that are involved in the server

- Motherboard
- CPU
- Memory

- Hard Drives
- Network Connection
- Power Supply

Key Difference between Hub, Switches, Routers, and Servers.

Features	Hub	Switch	Router	Server
Data Transmission	Broadcast to all Connected Devices	Directs data to correct port based on MAC address	Routes data between different networks using ip	Provide resources
Layer	Physical Layer	Data Link Layer	Network Layer	Application Layer
Traffic Management	No Management	Efficient Traffic Management	Efficient Routing	Manage Services
Security	No Security	Basic Security	Most Secure	Has Features
Addressing	No Addressing	MAC Address Table	Ip Addresses	Uses DNS and IPS
Ports	Typically Fewer	Many Ports	Fewer Ports	No Ports

Table 3

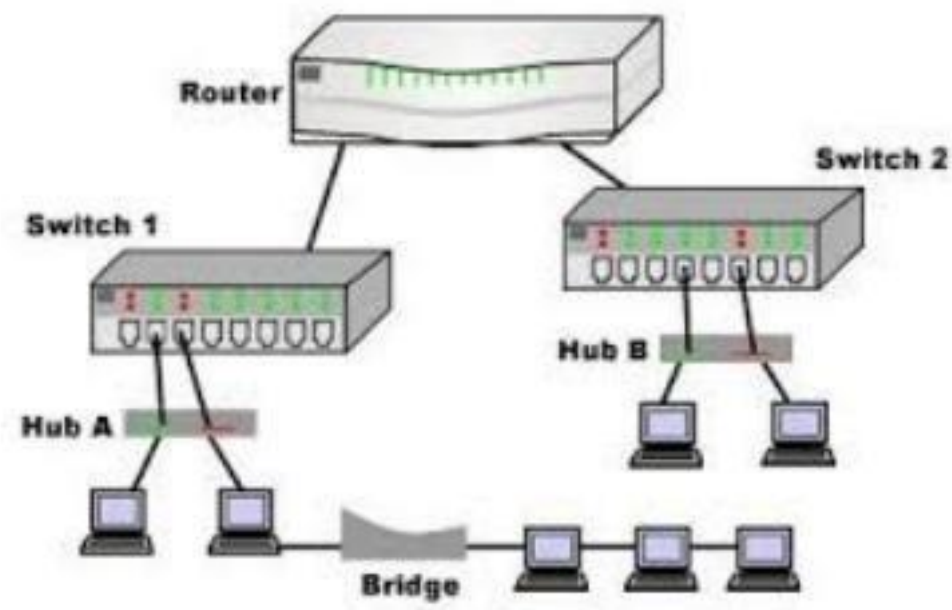


Figure 21

Lab3:

Experiment Name: Performing an Initial Switch and Router Configuration

Aim: Performing an Initial Configuration of a Cisco Catalyst 2960 switch

Apparatus (Software): Cisco Packet Tracer Description:

We have set the internet protocol address (IP) at PC0, PC1, PC2, PC3. Then the protocol will be prepared for required communication via a switch.

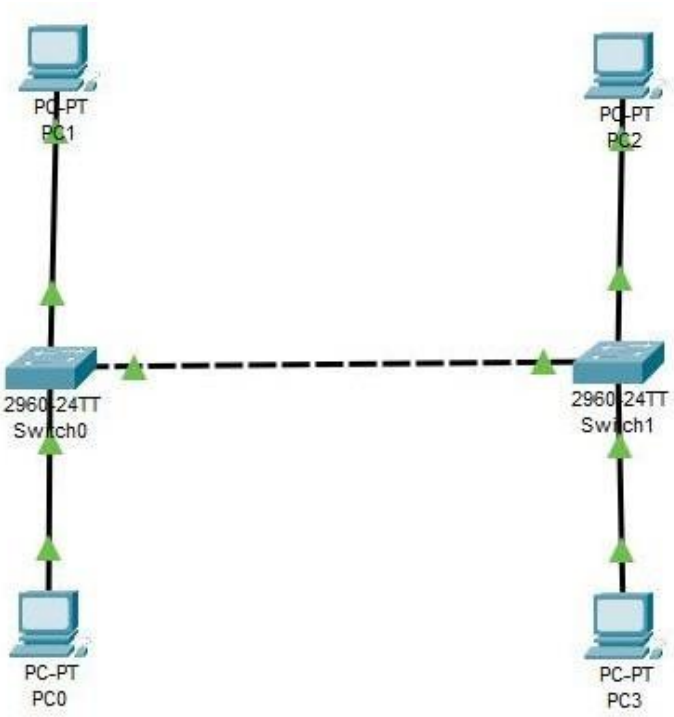


Figure 22

Each of the system has a unique IP address and properly connected to each other. Packet Sharing is also successful.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC1	PC0	ICMP		0.000	N	0	(edit)
	Successful	PC1	PC3	ICMP		0.000	N	1	(edit)
	Successful	PC1	PC2	ICMP		0.000	N	2	(edit)
	Successful	PC0	PC2	ICMP		0.000	N	3	(edit)
	Successful	PC0	PC3	ICMP		0.000	N	4	(edit)
	Successful	PC2	PC3	ICMP		0.000	N	5	(edit)

Figure 23

Router Configuration:

We have set the internet protocol address (IP) at PC0, PC1, PC2, PC3. Then the protocol will be prepared for required communication via a switch and have them attached to 2 routers. The network path created for them is also unique.

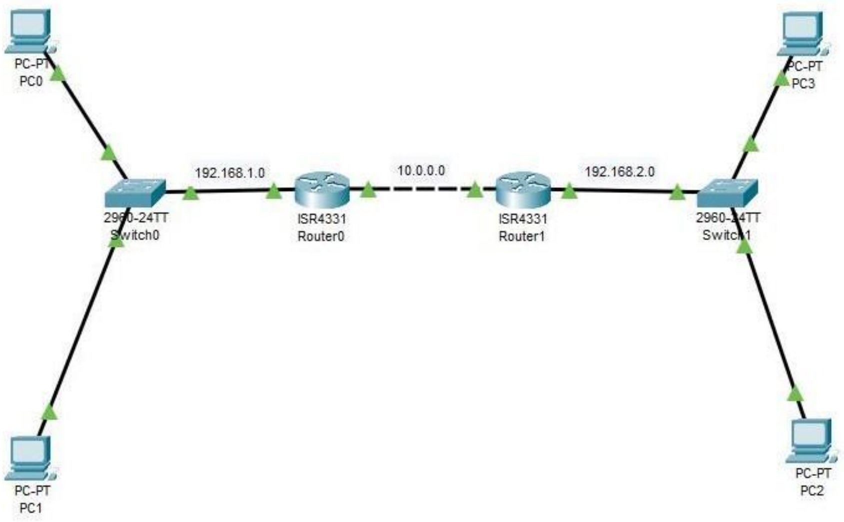


Figure 24

Each of the system has a unique IP address and are properly connected to each other. The routers are statically routed so they can communicate with each other.

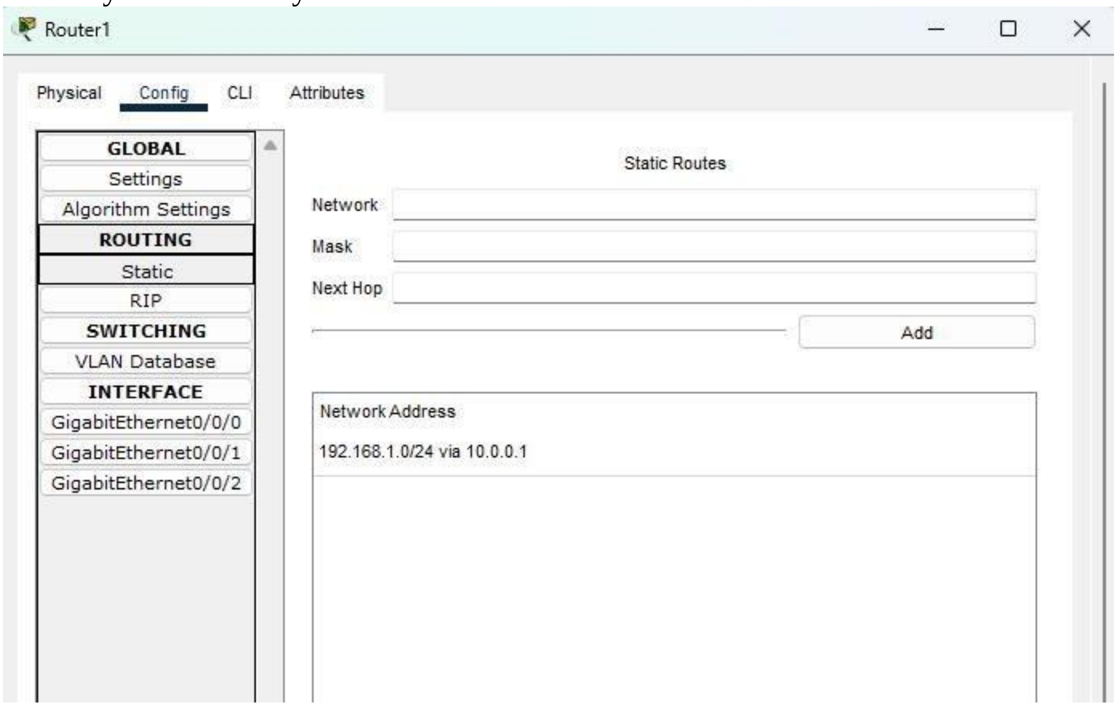


Figure 25

Lab4:

Experiment Name: Physical Topologies

Aim: Performing an configuration for the Physical Topologies to create a network.

Apparatus (Software): Cisco Packet Tracer.

Description:

We are going to perform Mesh, Star, Bus, Ring and Hybrid Topology to get how the physical network is used in real life. We will connect multiple systems to switches, hub, routers and achieve connectivity.

Mesh Topology:

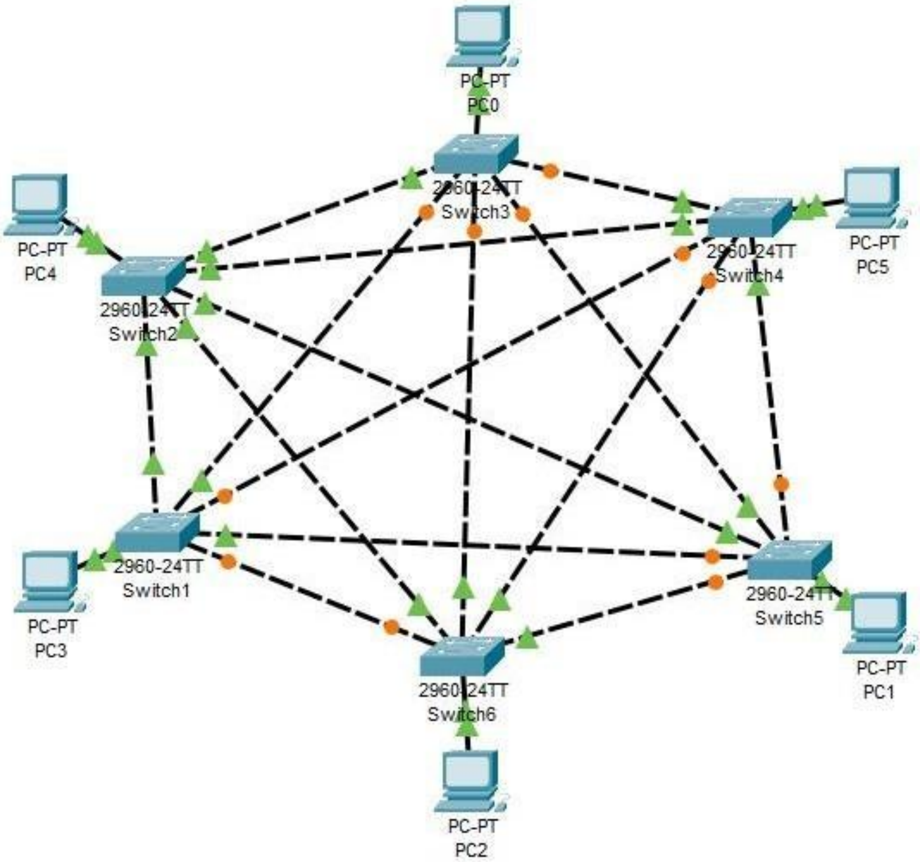


Figure 26

Each system has a unique IP address and connects to a dedicated switch, creating a point-to-point connection. Since a system has only one physical network port and cannot connect to multiple systems directly, a switch acts as a connector, enabling communication between multiple devices. The Packet Sharing is successful.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)
	Successful	PC0	PC5	ICMP		0.000	N	1	(edit)
	Successful	PC0	PC2	ICMP		0.000	N	2	(edit)
	Successful	PC0	PC3	ICMP		0.000	N	3	(edit)
	Successful	PC0	PC4	ICMP		0.000	N	4	(edit)

Figure 27

Star Topology:

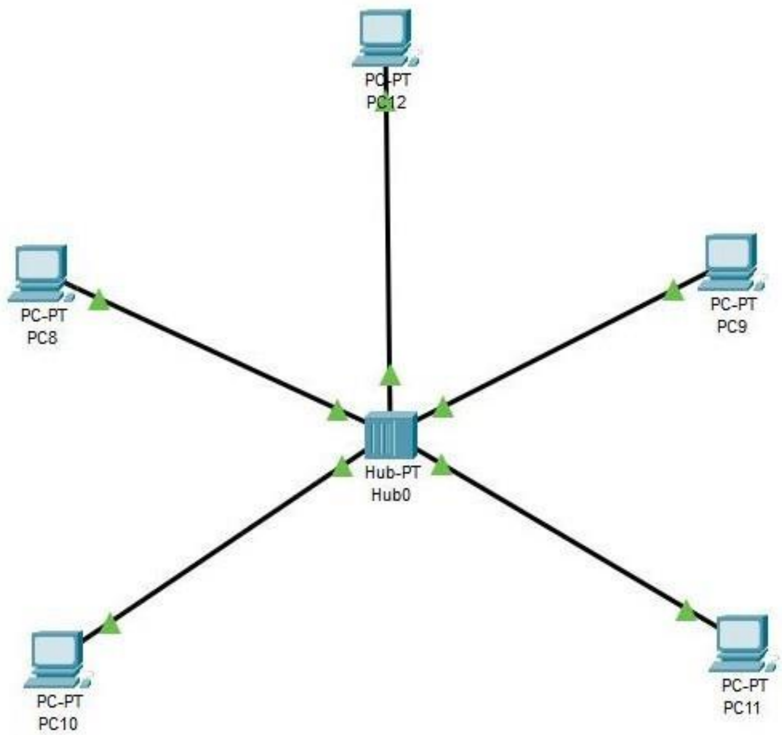


Figure 28

All the Systems are connected to a central Hub/switch, Creating a multi point connection. Each system has a unique IP address. The Packet Sharing is successful.

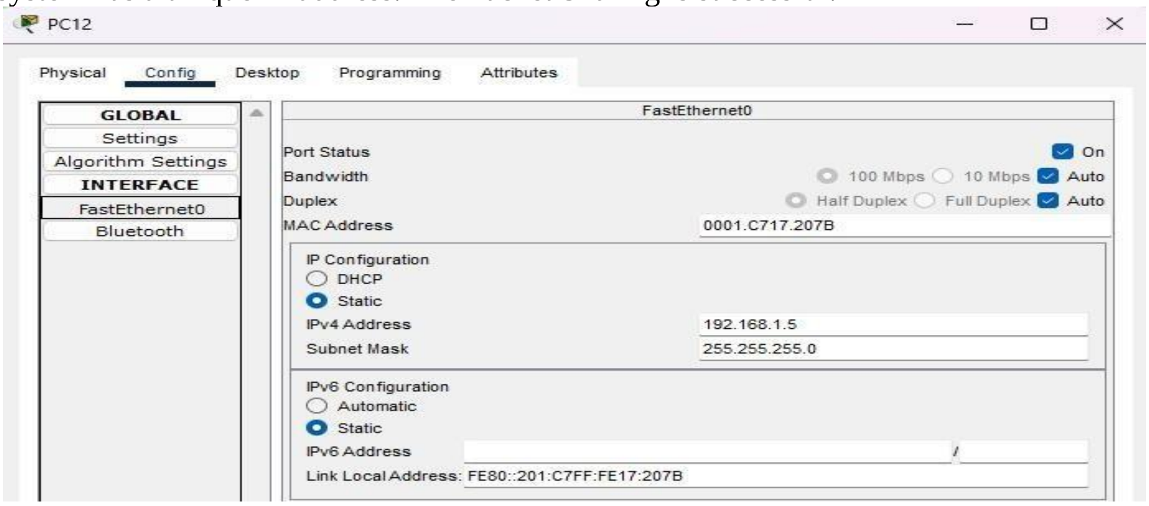
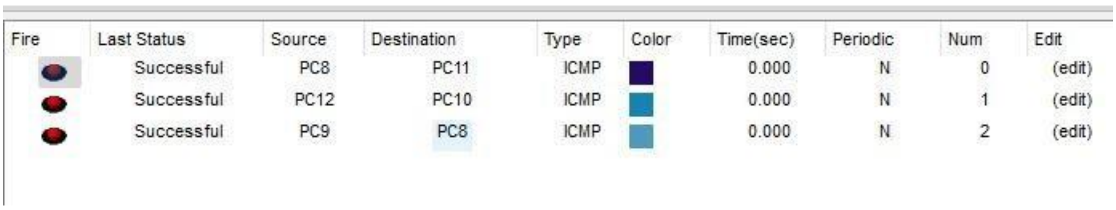


Figure 29



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC8	PC11	ICMP		0.000	N	0	(edit)
	Successful	PC12	PC10	ICMP		0.000	N	1	(edit)
	Successful	PC9	PC8	ICMP		0.000	N	2	(edit)

Figure 30

Bus Topology:

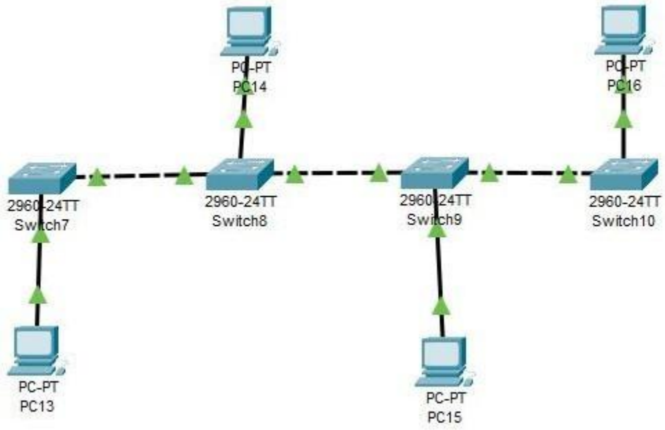


Figure 31

Each system has a unique IP address and connects to a shared communication medium, typically a single backbone cable, forming a bus topology. Since all devices share the same transmission channel, data travels along the backbone, and each system listens for packets addressed to it.

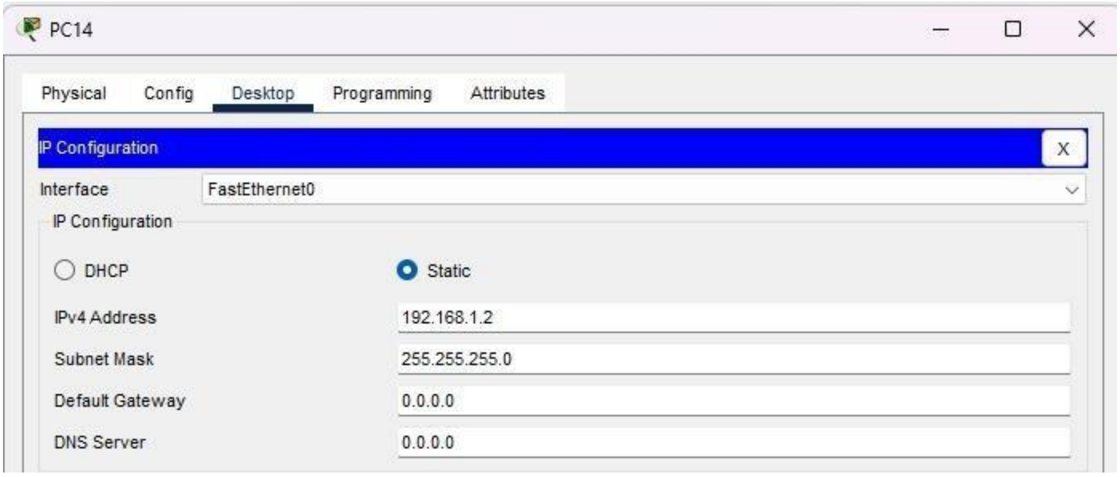


Figure 32

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC13	PC14	ICMP		0.000	N	0	(edit)
	Successful	PC13	PC15	ICMP		0.000	N	1	(edit)
	Successful	PC13	PC16	ICMP		0.000	N	2	(edit)

Figure 33

Ring Topology:

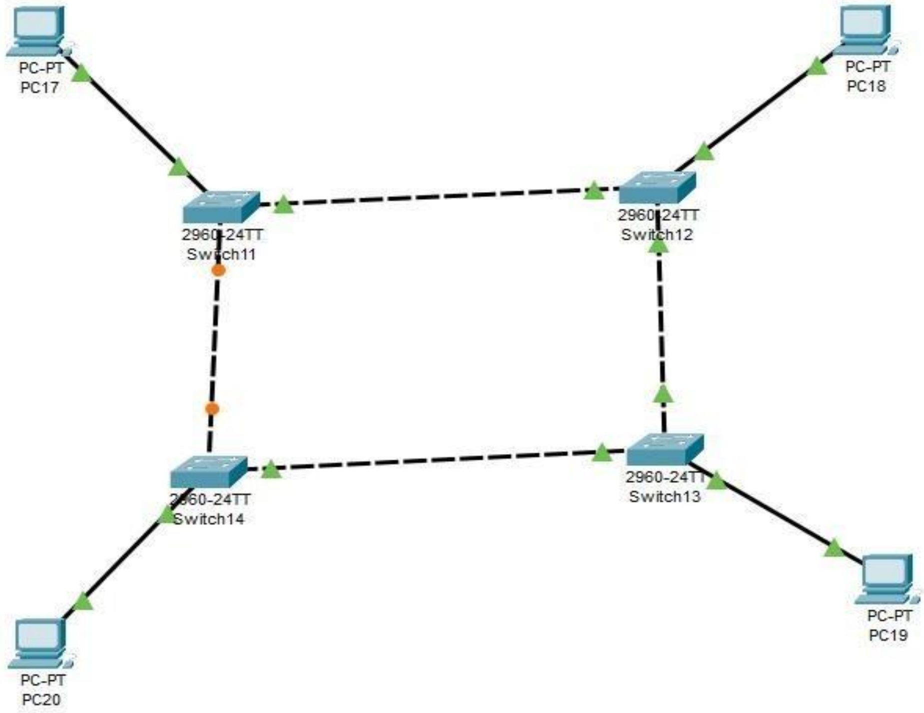


Figure 34

Each system has a unique IP address and connects to exactly two neighboring devices, forming a closed-loop structure known as a ring topology. Data travels in one or both directions around the ring until it reaches the intended destination, with each device acting as a repeater to maintain signal strength.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC17	PC20	ICMP		0.000	N	0	(edit)
	Successful	PC17	PC19	ICMP		0.000	N	1	(edit)
	Successful	PC17	PC18	ICMP		0.000	N	2	(edit)

Figure 35

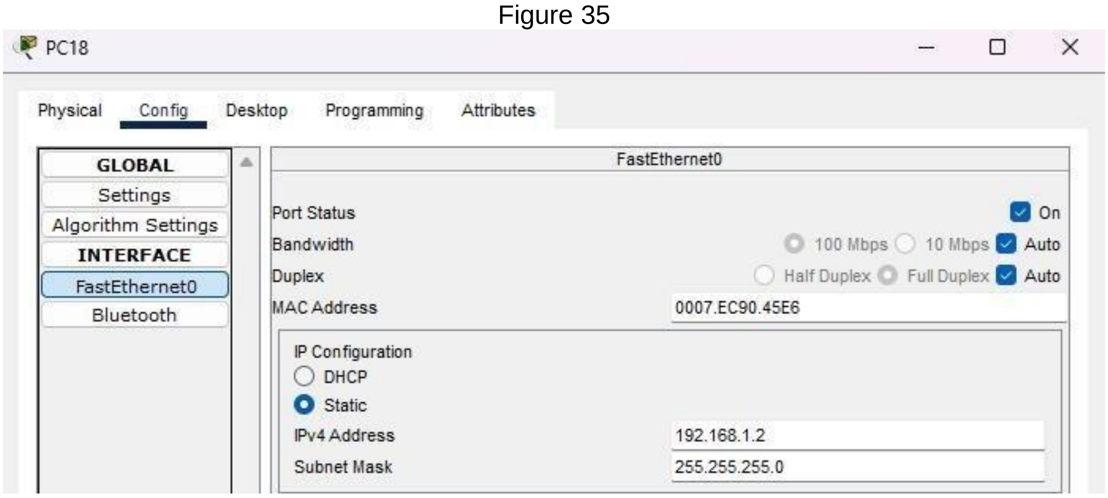


Figure 36

Hybrid Topology:

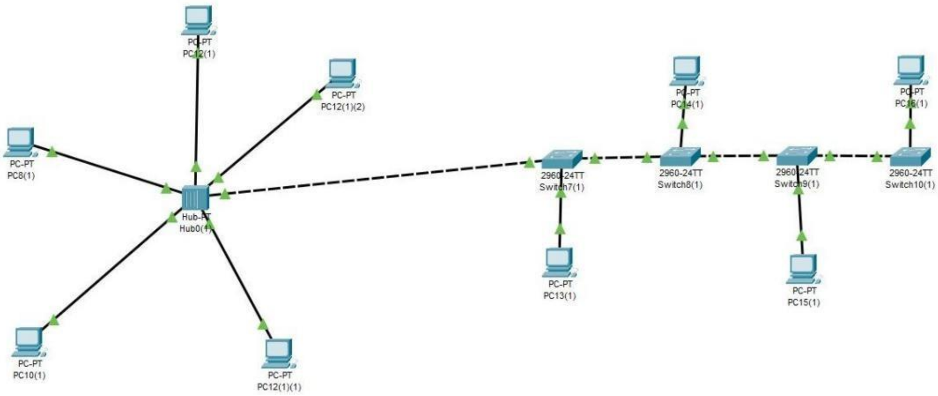


Figure 37

Each system has a unique IP address and connects using a combination of star and bus topologies, forming a hybrid network. In this setup, multiple devices connect to switches in a star topology, while these switches are interconnected in a bus-like manner, allowing efficient communication across the network while maintaining scalability and fault tolerance.

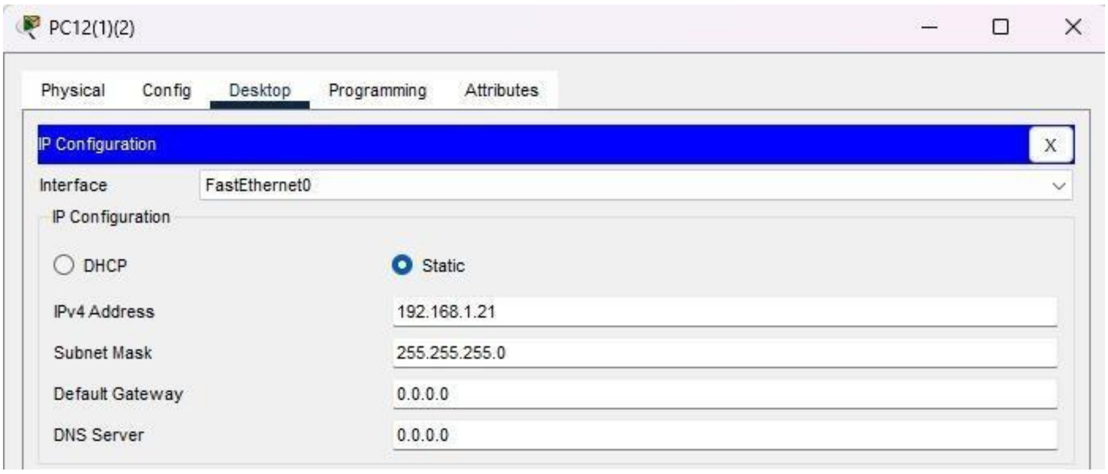


Figure 38

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC12(1)	PC13(1)	ICMP		0.000	N	0	(edit)

Figure 39

Packet is successfully shared as PC12(1) is in Star Network while PC13(1) is in Bus Topology.

Lab Quiz 1:

Question: Your university has three buildings: Admin Block, Computer Science Department, Library, HR, and Student Affairs. Each building has its internal network setup, and now the university wants to interconnect all three buildings for seamless communication.

- The Admin Block has a central connection where all computers share the same communication medium.
- The Computer Science Department follows a structure where each device is individually connected to a central device for efficient communication.
- The Library network is designed so that every computer is connected to multiple others, ensuring redundancy and reliability.
- HR Department - Operates in a way that computers are connected in a closed- loop, where data flows in a sequence, improving structured communication.
- Student Affairs uses a linear communication setup, where all devices are connected sequentially along a single cable, ensuring simple and cost-effective connectivity.

Your task is to design and implement the network for each building using different connection methods. Once completed, connect all five buildings into a unified network to enable smooth data transfer between them. Use appropriate devices and connections to ensure efficiency and reliability.

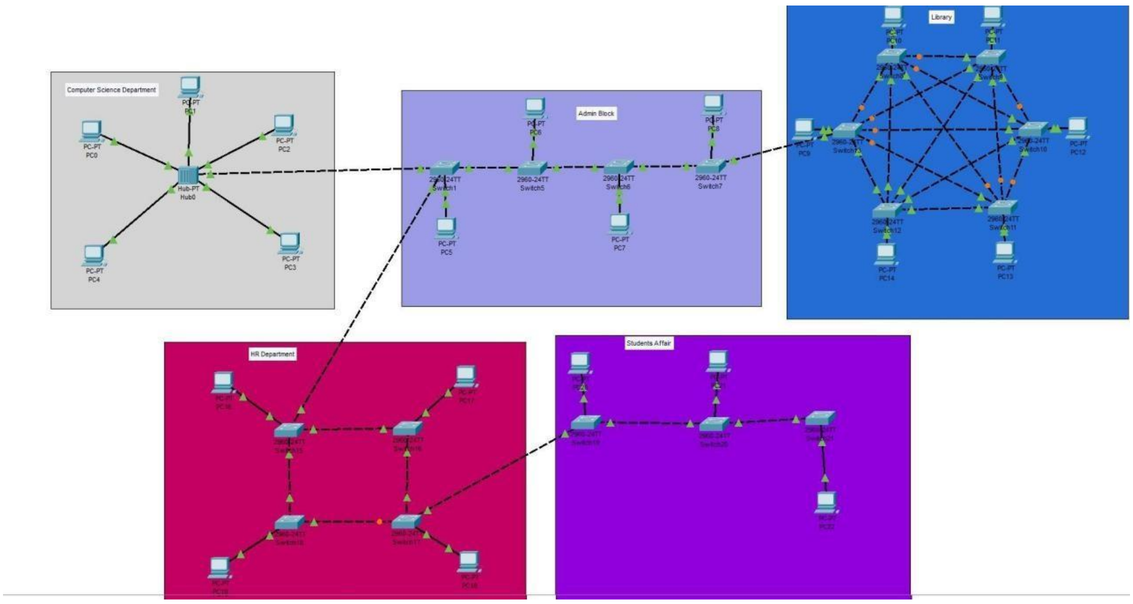


Figure 40

1. **Admin Block:** needs a Bus Topology
2. **Computer Science** Department: needs a Star Topology
3. **Library:** needs a Mesh Topology
4. **HR Department:** needs a Ring Topology
5. **Student Affairs:** needs a Bus Topology

Lab 5:

Experiment Name: Router Configuration.

Aim: Configuring routers

Apparatus (Software): Cisco Packet Tracer

Descriptions:

We will configure routers manually as well as using CLI. As in previous lab we got the idea related to how to configure the router but today we will discuss it in detail. To start we have a simple setup of 4 Systems connected to 2 switches and then are attached to a single router.

Configuration Default Method:

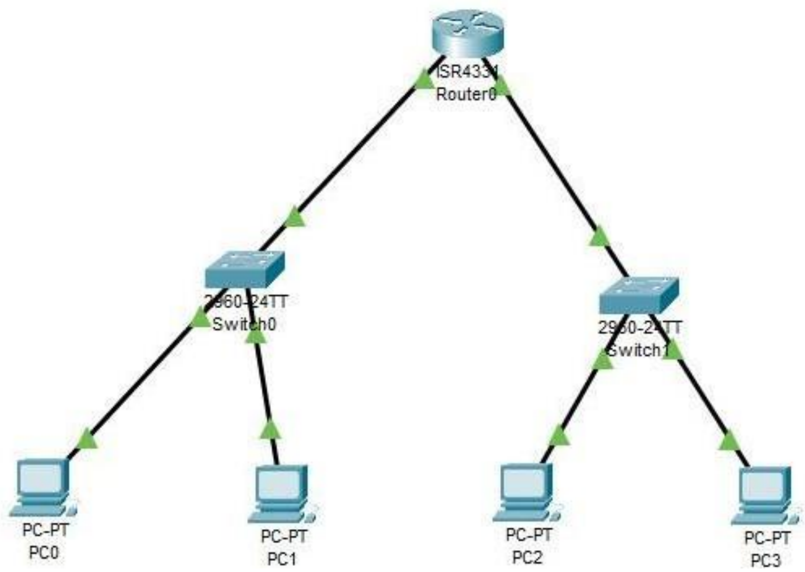


Figure 41

The following figure depicts how the settings of router looks like.

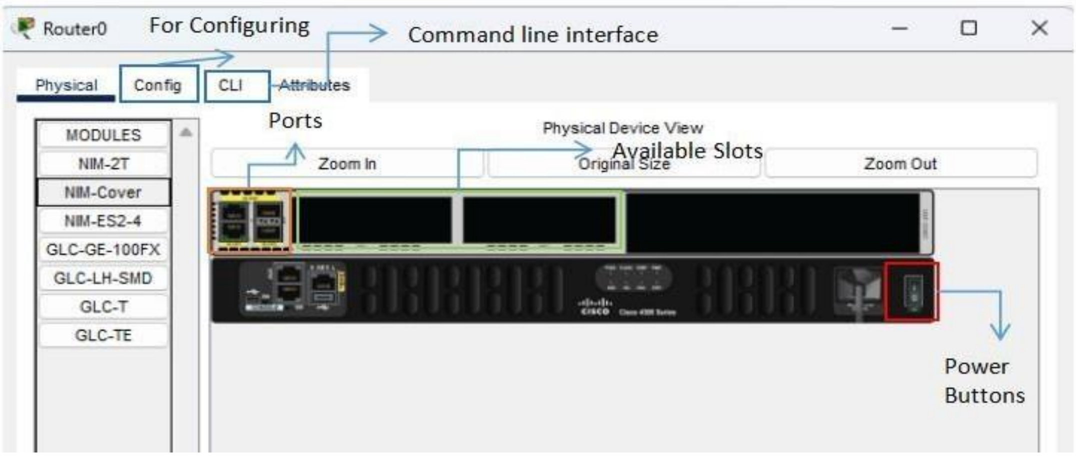


Figure 42

In the previous figure we saw 4 tabs also some buttons and ports. The following figure depicts all the configuration possible.

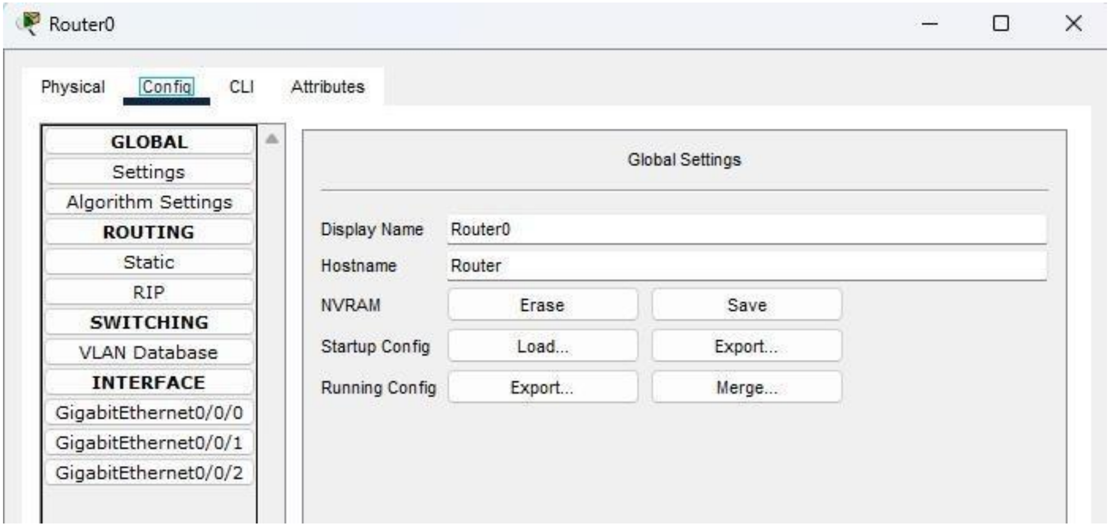


Figure 43

As you are able to observe that in the interface tab we only have 3 ports they are expandable and each port is designed to create a network. Those networks become the default network gateway so each network have some routing possibility.

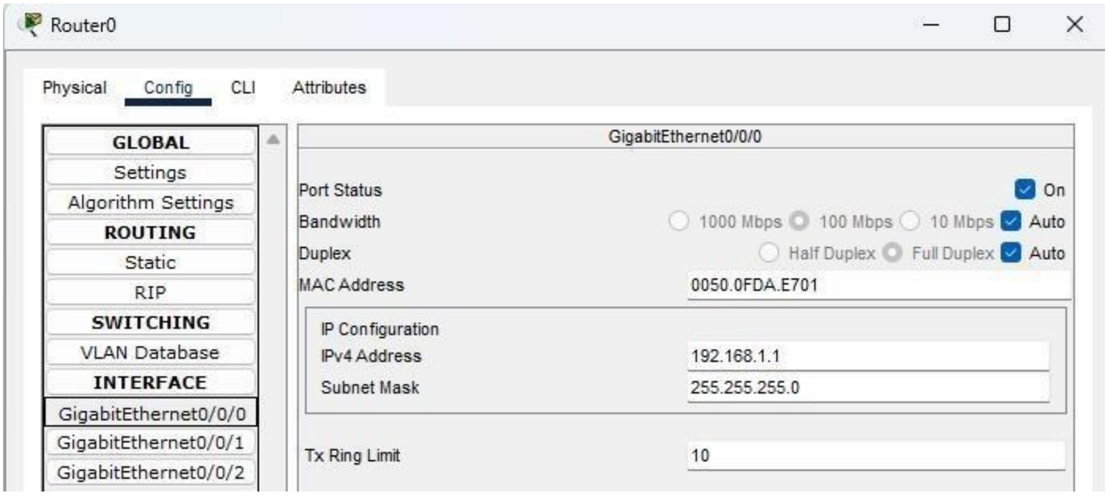


Figure 44

the router has 3 Gigabit Ethernet ports, each assigned a specific gateway IP address. These gateways enable the router to efficiently route packets between different network segments, allowing seamless communication between connected devices.

Configuration using CLI:
First of all select the CLI tab

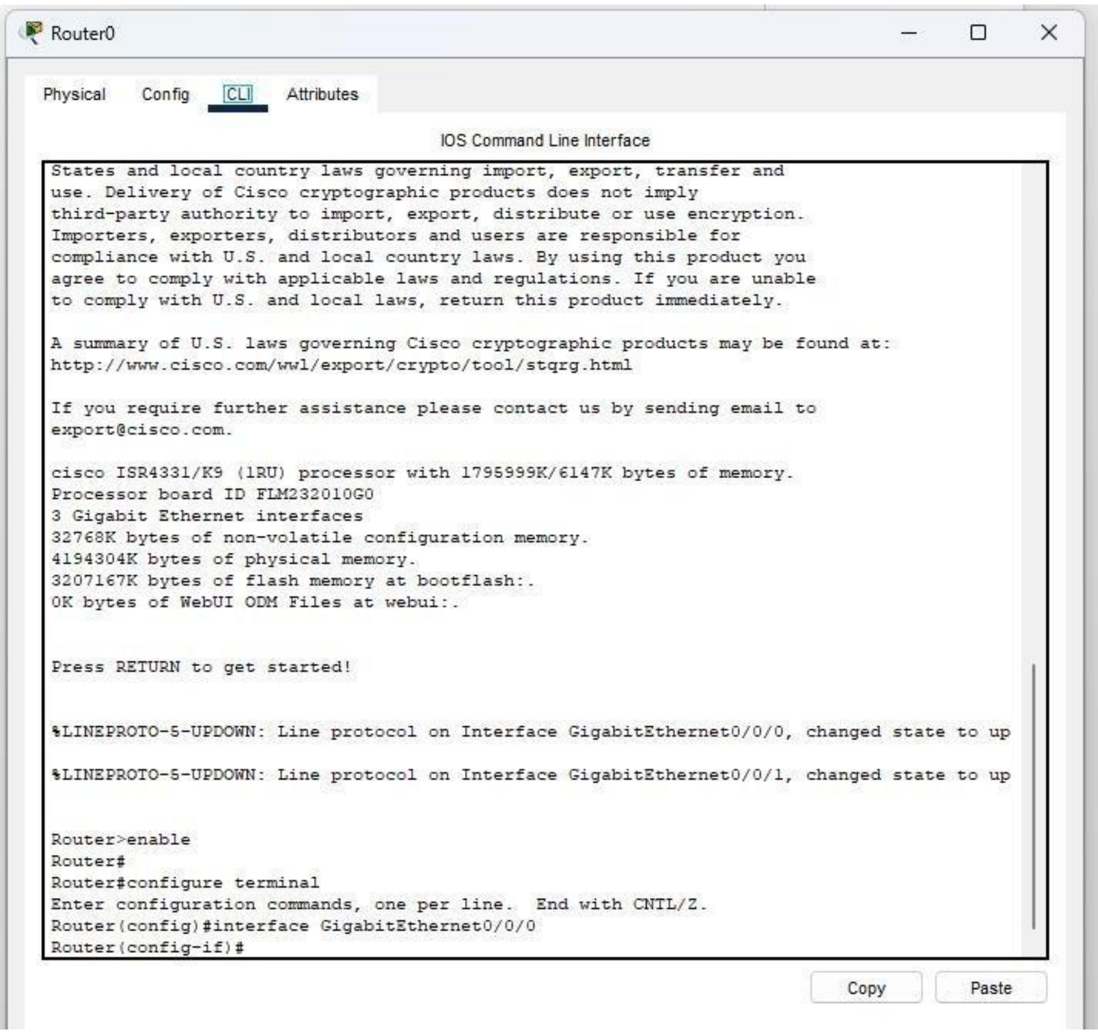


Figure 45

Enter privileged EXEC mode:

```
Router>enable
```

Copy

Figure 46

Enter Global Configuration mode:

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
```

Figure 47

Configure Port g0/0/0 to Switch 0 :

```
Router(config)#interface g0/0/0
Router(config-if)#ip add 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
exit
```

Figure 48

Configure Port g0/0/1 connected to Switch 1:

```
Router(config)#interface g0/0/1
Router(config-if)#ip add 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
exit
```

Figure 49

Lab 6: Part 1

Experiment Name: Static Routing

Aim: Making it possible for multiple networks to connect and share messages.

Apparatus (Software): Cisco Packet Tracer

Description: Static routing is a manual method of defining network paths in a router's routing table.

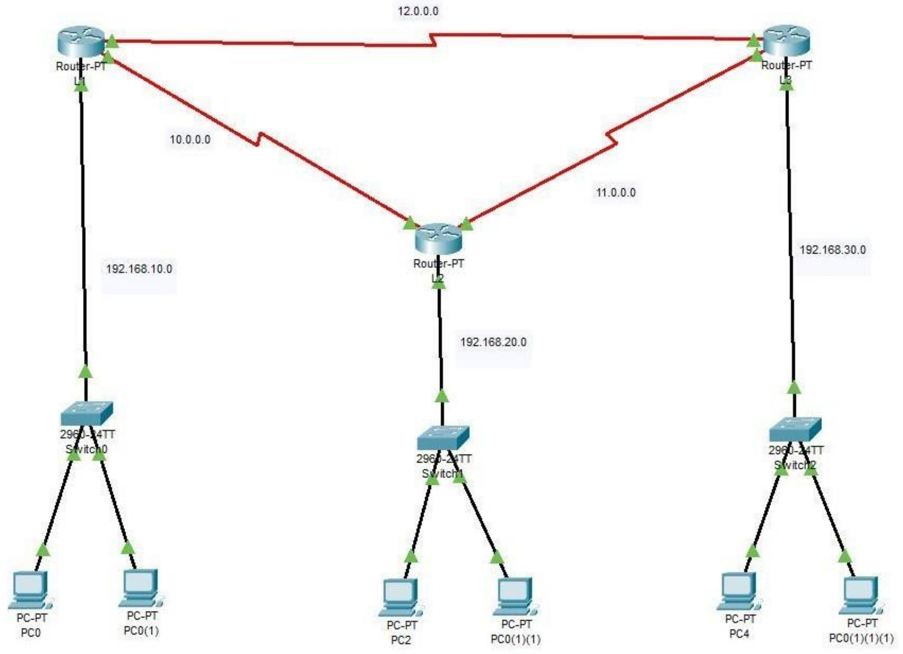


Figure 50

Lets take the above figure as an example. There are 3 routers which are connected using serial cable. Each router has its own network. Our objective is to make it possible that they communicate with each other.

L1 router has its own Local Network consists of 192.168.10.0 L2 router has its own Local Network consists of 192.168.20.0 L3 router has its own Local Network consists of 192.168.30.0

While it uses 10.0.0.0, 11.0.0.0, 12.0.0.0 for inter router communication.

To perform inter router communication we need to correctly perform static routing which is like of deciding hops, which is basically deciding routes for the data to go. It can be done manually and using CLI as well.

Also we add a New Module so it can accommodate the port to connect.

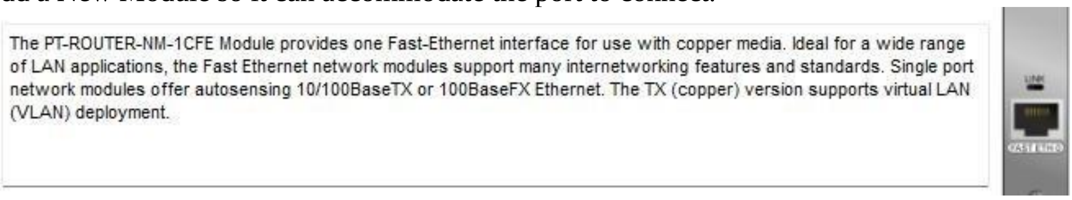


Figure 51

Manually:

Lets take the above figure as an example and understand how the hoping works

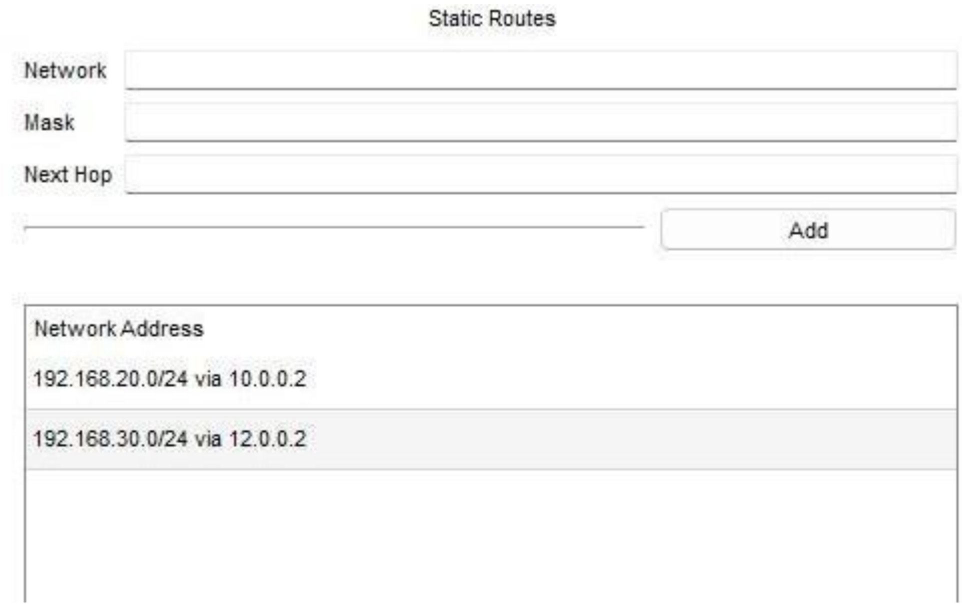


Figure 52

To start things of hops, require the HOST network, sub-net, and the IP address of the Destination Router (in our case, there are 2 routers with networks 10.0.0.0 and 12.0.0.0).
Given in the figure we have set two hops. First will give Data to L2 router and the second hop takes Data to L3 router.
You have to manually set these hops on the other routers as well. The packets are delivered successfully
How the Hope will Work:

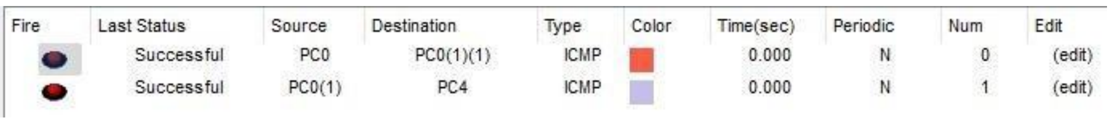
PC0 to PC0(1)(1)

Source: PC0 **Destination:** PC0(1)(1) **Packet Type:** ICMP
Status: Successful (The ping was received, meaning the static routes are working.)

Time Taken: 0.000 sec PC0(1) to PC4

Source: PC0(1) **Destination:** PC4 **Packet Type:** ICMP
Status: Successful (PC0(1) can communicate with PC4.)

Time Taken: 0.000 sec



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
	Successful	PC0	PC0(1)(1)	ICMP		0.000	N	0	(edit)
	Successful	PC0(1)	PC4	ICMP		0.000	N	1	(edit)

Figure 53

Using CLI:
Assign IP Addresses to Router Interfaces:
Router1 (L1):

```
Router>en
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#ip add 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
```

Figure 54

```
Router(config-if)#interface se2/0
Router(config-if)#ip add 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
```

Figure 55

```
Router(config)#interface se3/0
Router(config-if)#ip add 12.0.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
exit
```

Figure 56

Router2 (L2):

```
Router>en
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip add 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to u
exit
```

Figure 57

```
Router(config)#interface se3/0
Router(config-if)#ip add 10.0.0.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
exit
```

Figure 58

```
Router(config-if)#interface se2/0
Router(config-if)#ip add 11.0.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
```

Figure 59

Router3 (L3):

```
Router>en
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip add 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
```

Figure 60

```
Router(config)#interface se3/0
Router(config-if)#ip add 12.0.0.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
exit
```

Figure 61

```
Router>en
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface se2/0
Router(config-if)#11.0.0.2 255.255.255.0
                  ^
% Invalid input detected at '^' marker.

Router(config-if)#ip add 11.0.0.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
```

Figure 62

Configure Static Routes
Router1 (L1):

```
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.30.0 255.255.255.0 12.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
```

Figure 63

Router2 (L2):

```
Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router(config)#ip route 192.168.30.0 255.255.255.0 11.0.0.2
Router(config)#no shutdown
                   ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
```

Figure 64

Router3 (L3):

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.10.1 255.255.255.0 12.0.0.1
%Inconsistent address and mask
Router(config)#ip route 192.168.10.0 255.255.255.0 12.0.0.1
Router(config)#ip route 192.168.20.0 255.255.255.0 11.0.0.1
Router(config)#no shutdown
                   ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
```

Figure 65

Lab 06: Part 2

Experiment Name: Dynamic Routing

Aim: Making it possible for multiple networks to connect and share messages.

Apparatus (Software): Cisco Packet Tracer

Description: To configure and verify dynamic routing using the Routing Information Protocol (RIP) in Cisco Packet Tracer.

Using CLI:

Router 1:

```
R1>en
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip add 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#int se2/0
R1(config-if)#ip add 10.0.0.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#int se3/0
R1(config-if)#ip add 12.0.0.2 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#network 192.168.10.0
R1(config-router)#network 10.0.0.0
R1(config-router)#network 12.0.0.0
R1(config-router)#exit
R1(config)#do wr
Building configuration...
[OK]
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 66

Router 2:

```
Router>en
Router#configure t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#int f0/0
R2(config-if)#ip add 192.168.20.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#int se2/0
R2(config-if)#ip add 11.0.0.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#int se3/0
R2(config-if)#ip add 10.0.0.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#no auto-summary
R2(config-router)#network 192.168.20.0
R2(config-router)#network 10.0.0.0
R2(config-router)#network 11.0.0.0
R2(config-router)#exit
R2(config)#do wr
Building configuration...
[OK]
R2(config)#^Z
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 67

Router 3:

```
Router>en
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#int f0/0
R3(config-if)#ip add 192.168.30.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#int se2/0
R3(config-if)#ip add 11.0.0.2 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#int se3/0
R3(config-if)#ip add 12.0.0.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#router rip
R3(config-router)#no auto-summary
R3(config-router)#192.168.30.0
                  ^
% Invalid input detected at '^' marker.

R3(config-router)#network 192.168.30.0
R3(config-router)#network 11.0.0.0
R3(config-router)#network 12.0.0.0
R3(config-router)#exit
R3(config)#do wr
Building configuration...
[OK]
R3(config)#
```

Figure 68

Lab 11: Part 1.

Experiment Name: Server Application

Aim: DNS Routing and accessing the DNS.

Apparatus (Software): Cisco Packet Tracer

Description: Using the Browser App and access the DNS assigned to the Server.

Connection Diagram:

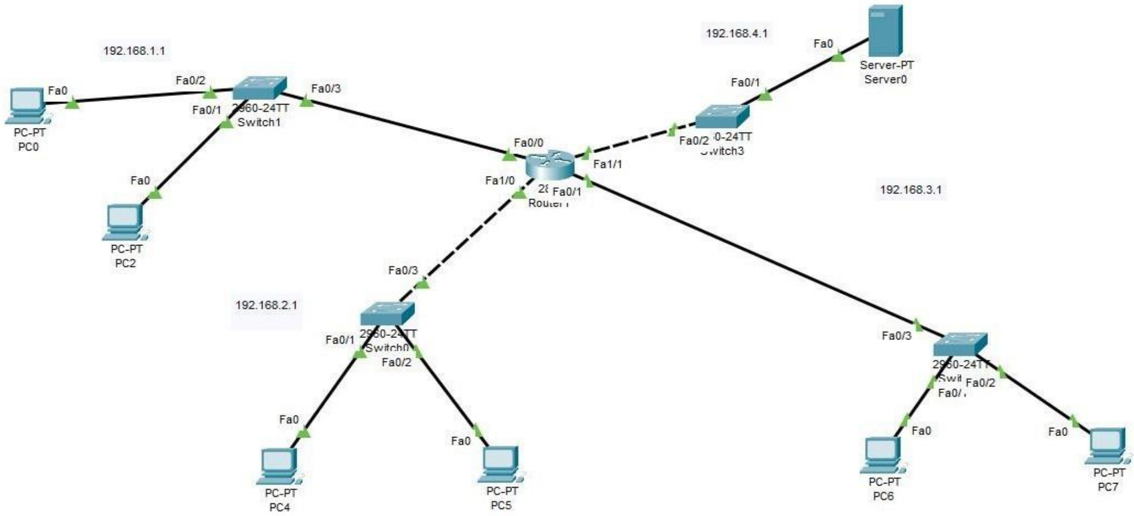


Figure 69

Layout Explanation:

This network diagram illustrates a three-route connection to a dedicated server. Each route is configured to provide reliable access to the server, which has a unique IP address assigned. This DNS serves as the dedicated link used to access the server and the hosted URL.

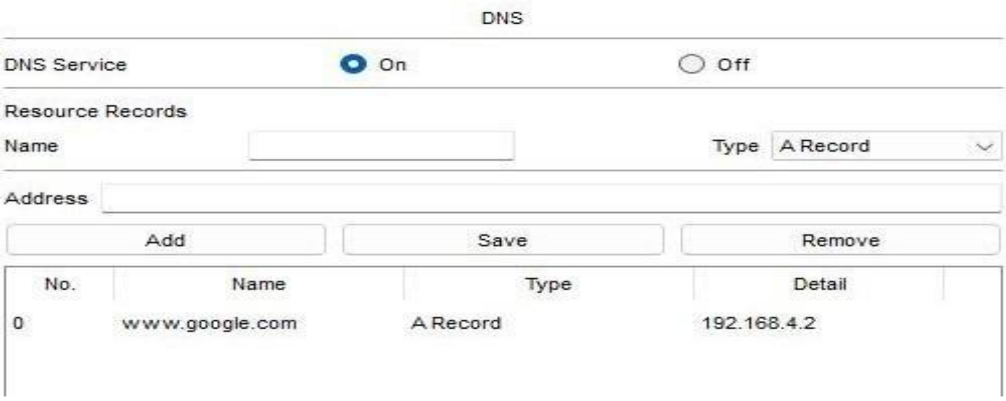


Figure 70

Using PC0 To access the Specific DNS.

Figure 71

After Accessing the DNS we get the following Result.

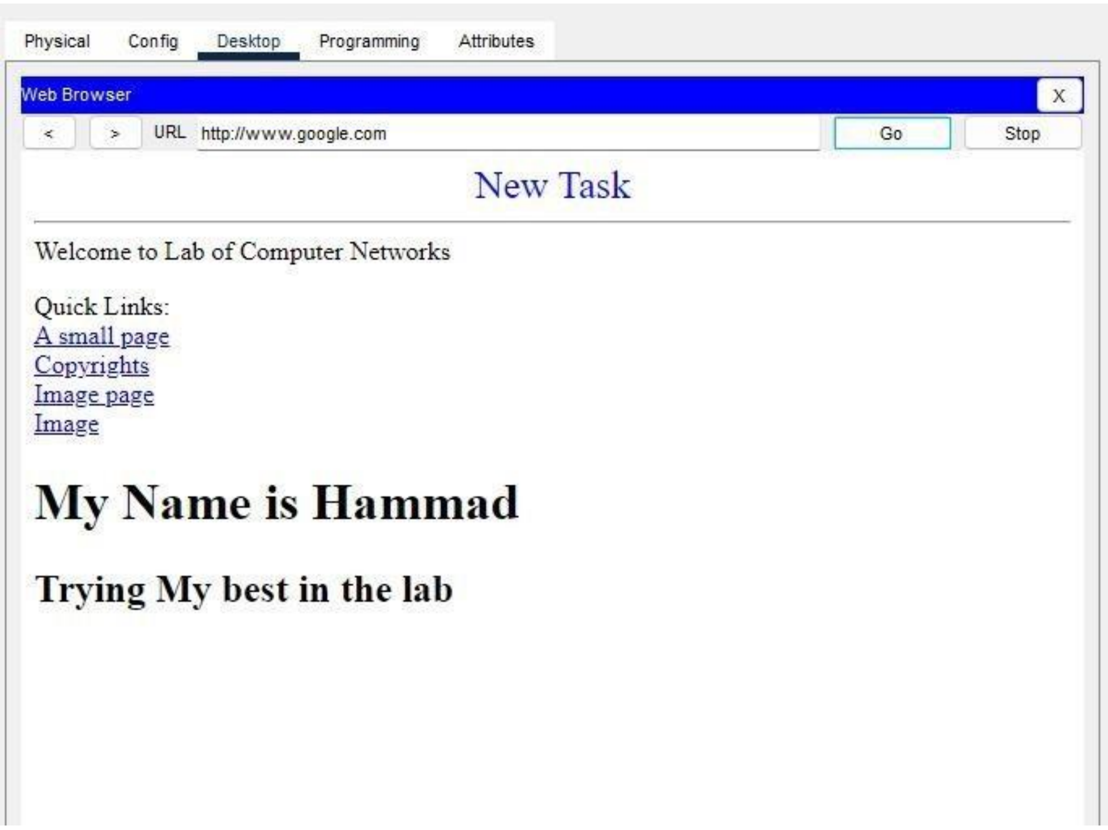


Figure 72

Lab 11: Part 2.

Experiment Name: **DCHP Protocol.**

Aim: Dynamic Allocation of IP’s to the System using CMD.

Apparatus (Software): **Cisco Packet Tracer**

Description: Dynamic Allocation of systems instead of Static Addressing using CMD.

Connection Diagram:

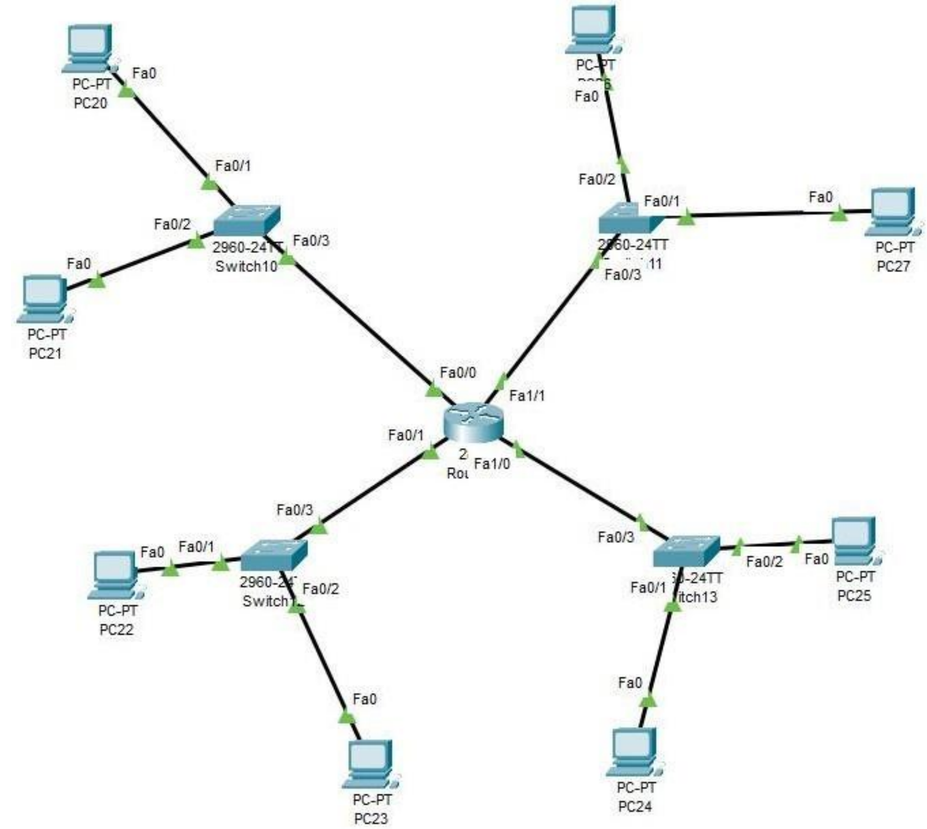


Figure 73

Layout Explanation:

Each and every node is designed to automatically get assigned IP in the system. The system has a gateway and the addressing is done according to the gateway which helps identify routing and makes it easier to connect with other nodes.

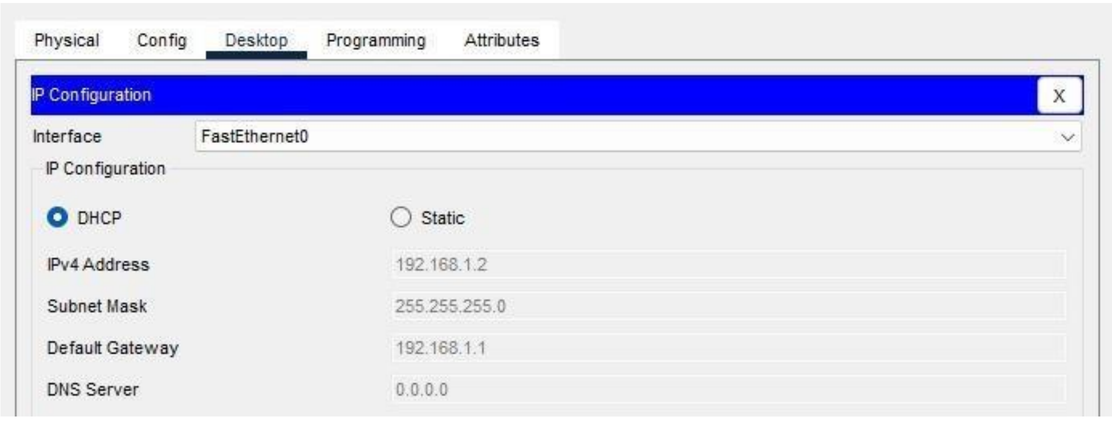


Figure 74

.

Each port on the router helps in the assigning process.

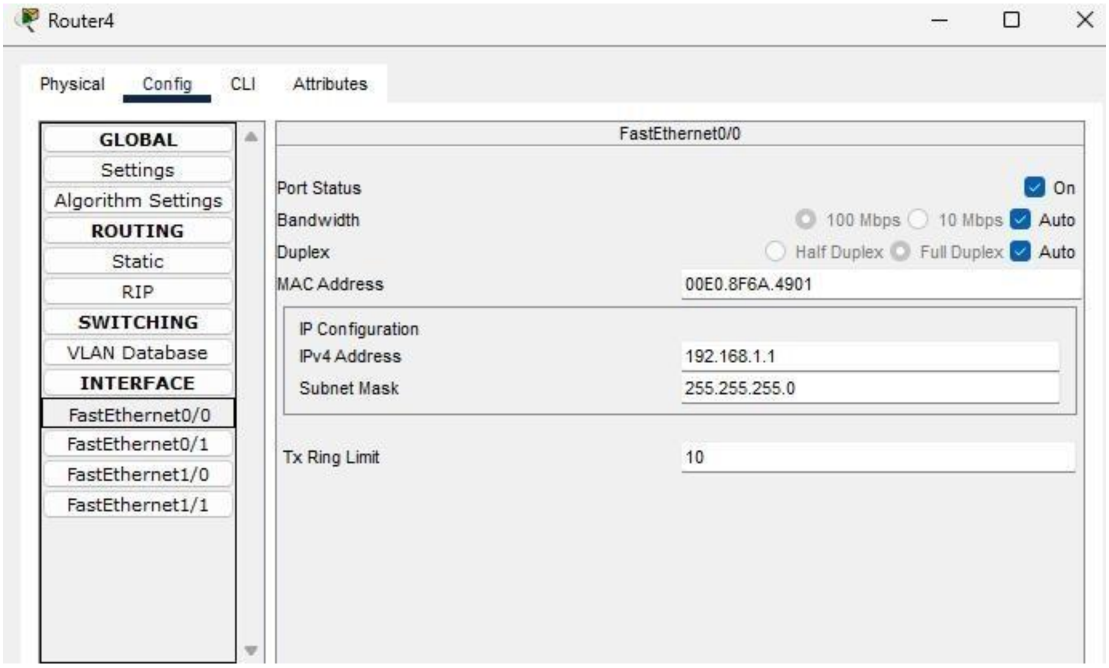


Figure 75

With the this we can easily assign and connect with the systems. As given below.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC21	PC24	ICMP		0.000	N	0	(edit)	

Figure 76

Using CLI to assign.

```
enable
configure terminal
! Interface Fa0/0 - Subnet 192.168.1.0/24
interface FastEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
ip dhcp pool VLAN1
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8
! Interface Fa0/1 - Subnet 192.168.2.0/24
interface FastEthernet0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
```

```
exit
ip dhcp pool VLAN2
network 192.168.2.0 255.255.255.0
default-router 192.168.2.1
dns-server 8.8.8.8
! Interface Fa1/0 - Subnet 192.168.3.0/24
interface FastEthernet1/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
ip dhcp pool VLAN3
network 192.168.3.0 255.255.255.0
default-router 192.168.3.1
dns-server 8.8.8.8
! Interface Fa1/1 - Subnet 192.168.4.0/24
interface FastEthernet1/1
ip address 192.168.4.1 255.255.255.0
no shutdown
exit
ip dhcp pool VLAN4
network 192.168.4.0 255.255.255.0
default-router 192.168.4.1
dns-server 8.8.8.8
! Exclude router IPs from being assigned
ip dhcp excluded-address 192.168.1.1
ip dhcp excluded-address 192.168.2.1
ip dhcp excluded-address 192.168.3.1
ip dhcp excluded-address 192.168.4.1
end
write memory
```

Lab 11: Part 3.

Experiment Name: DCHP Protocol.

Aim: Making IP Pool.

Apparatus (Software): Cisco Packet Tracer

Description: IP Pool to assign IPs dynamically to the nodes.

Connection Diagram.

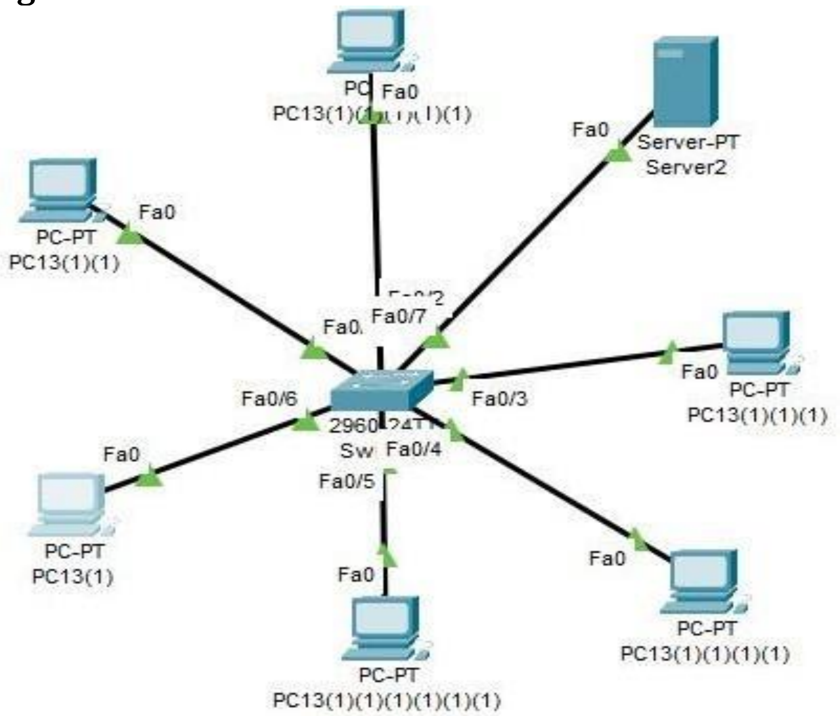


Figure 77

The Server has a Pool to assign IP directly to nodes using Server.

You will Start the Service and then assign a proper gateway to all the nodes connected to the server.

Then we will assign the starting IP and then we will save them

A Starting point is set and u can adjust how many user can connect to the server.

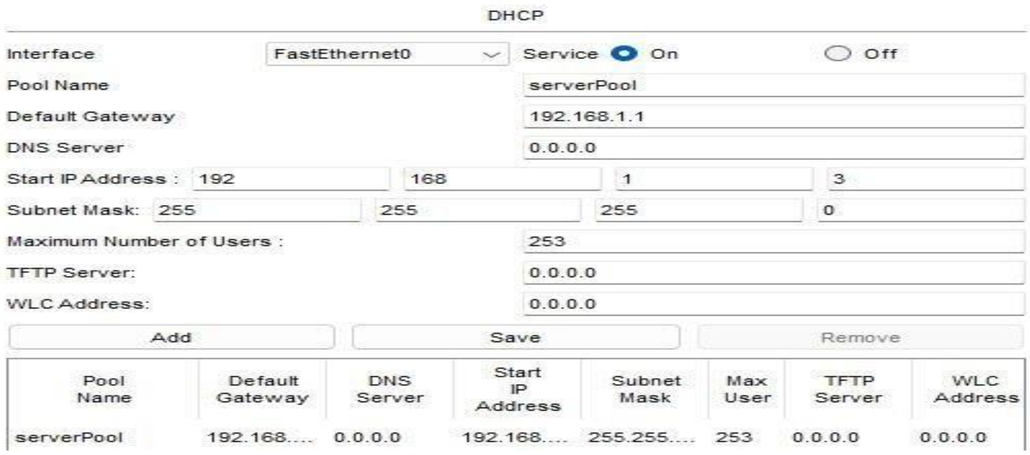


Figure 78

The Node will dynamic Pick IP from the IP pool.

Figure 79

Lab 11: Part 4.

Experiment Name: DCHP Protocol. Aim: Wireless Mapping.

Apparatus (Software): Cisco Packet Tracer

Description: Creating a Wireless Connection to connect to multiple Nodes.

Connection Diagram:

Figure 80

Layout Description:

One Wireless Router is connected to multiple node with wifi modules which helps the nodes to detect the wireless signals coming out of the router.

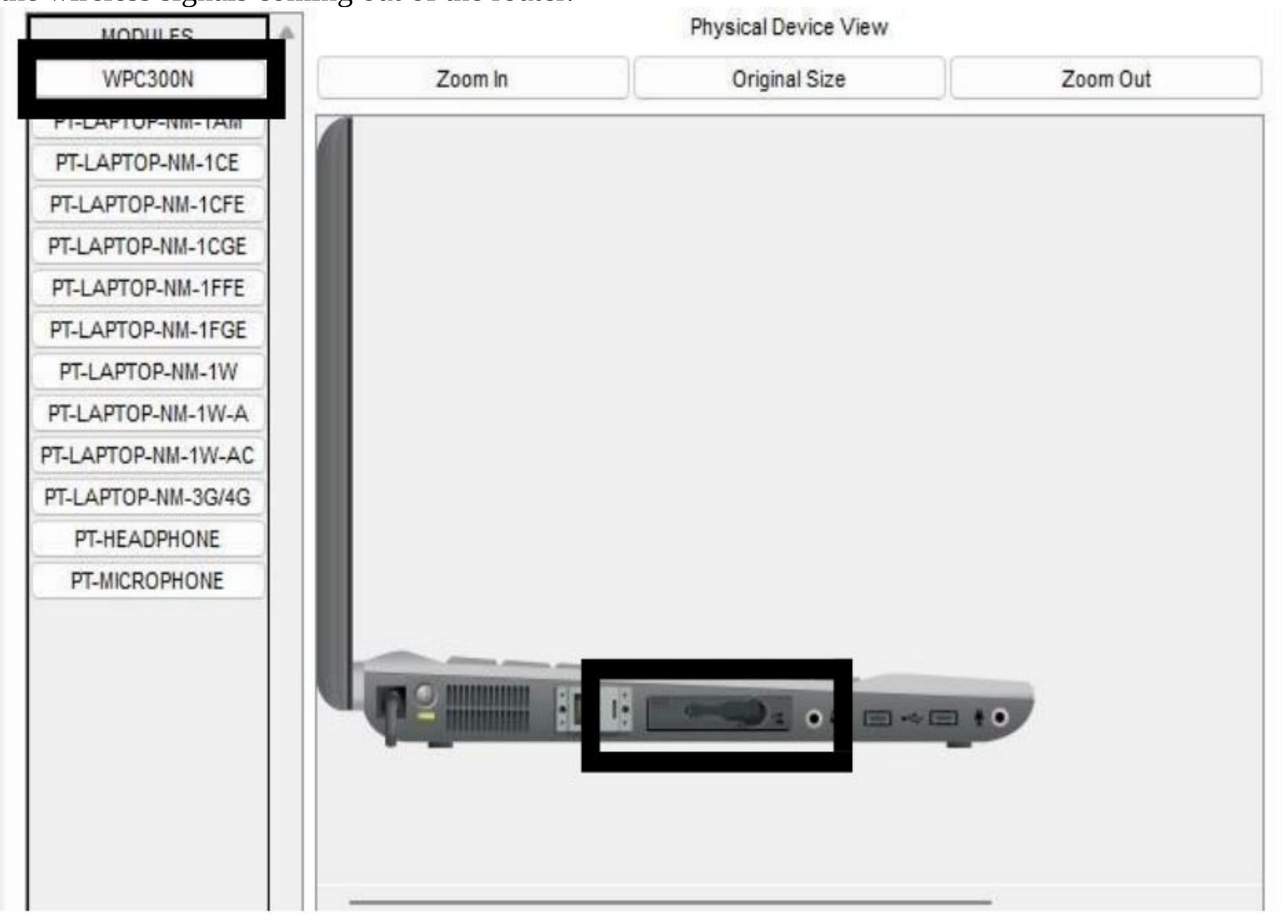


Figure 81

After replacing the module to Wifi we are able to see the router in the connection tab.

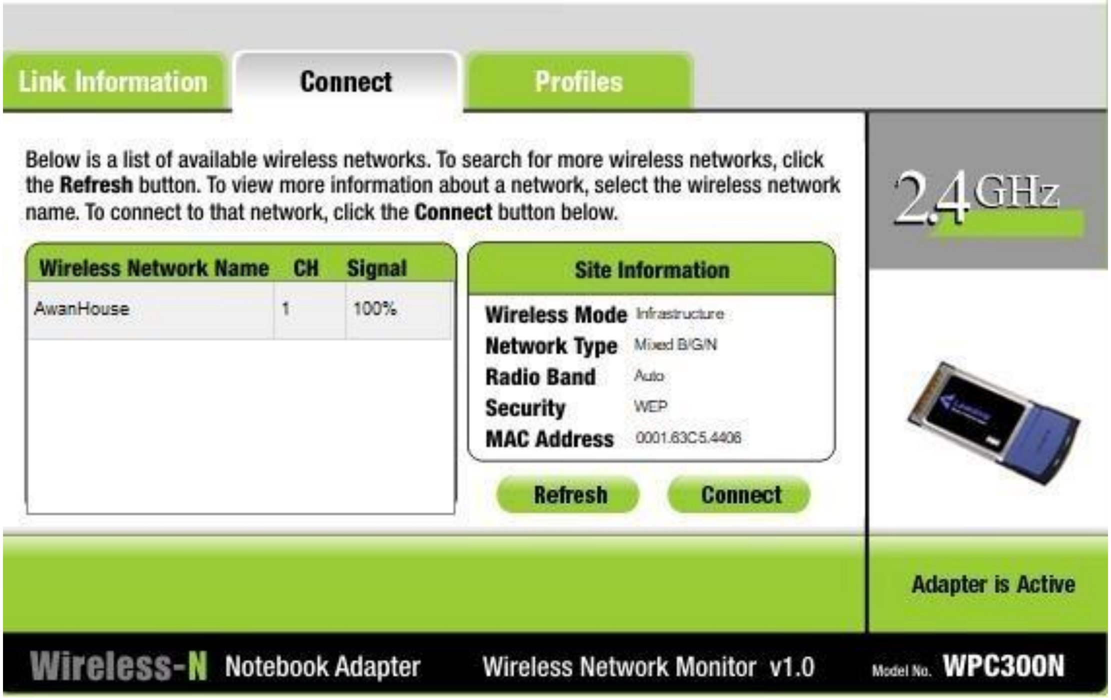


Figure 82

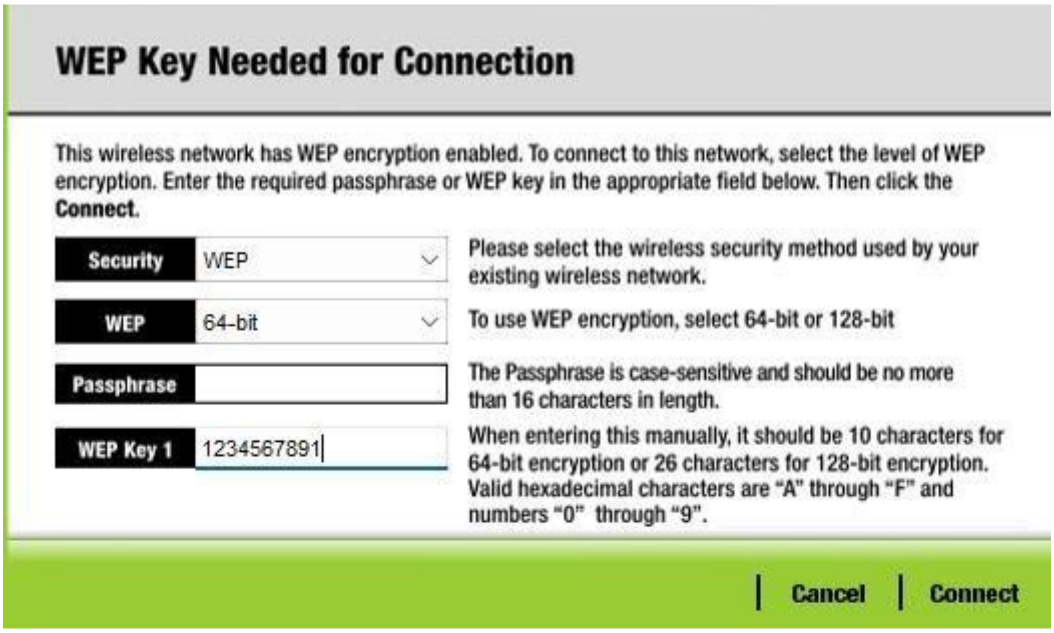


Figure 83

The WEP key is the security key that we assign to the router with which we connect the router.

Accessing the router Default Page on the user side.

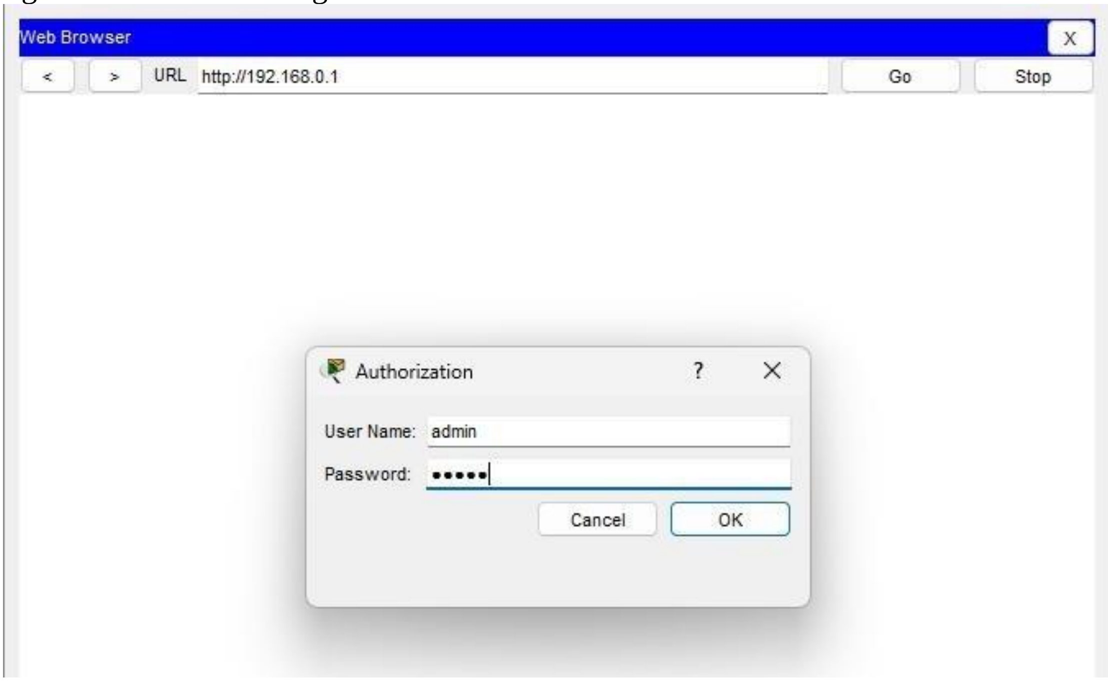


Figure 84

You can change the settings of the Router in this section.

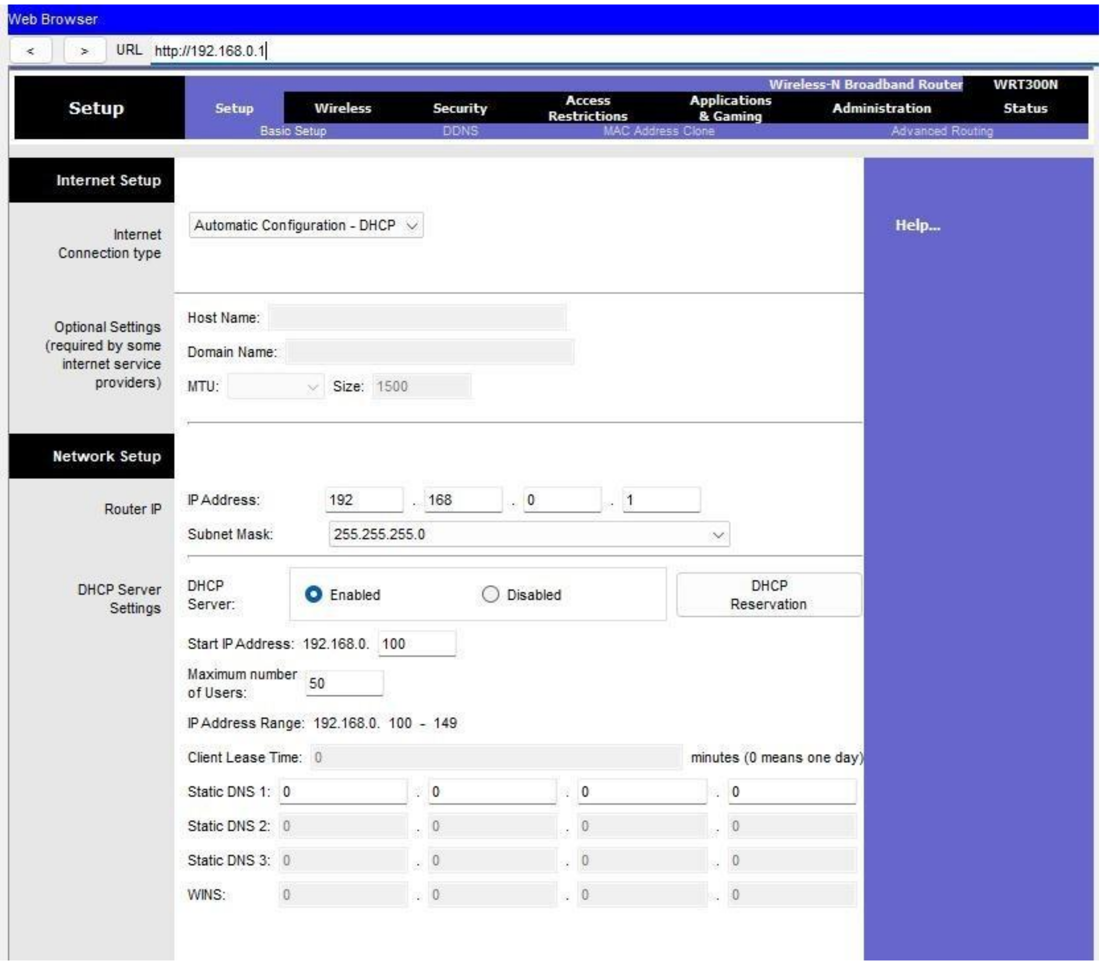


Figure 85

On router Side you can access the router default page on the GUI tab.

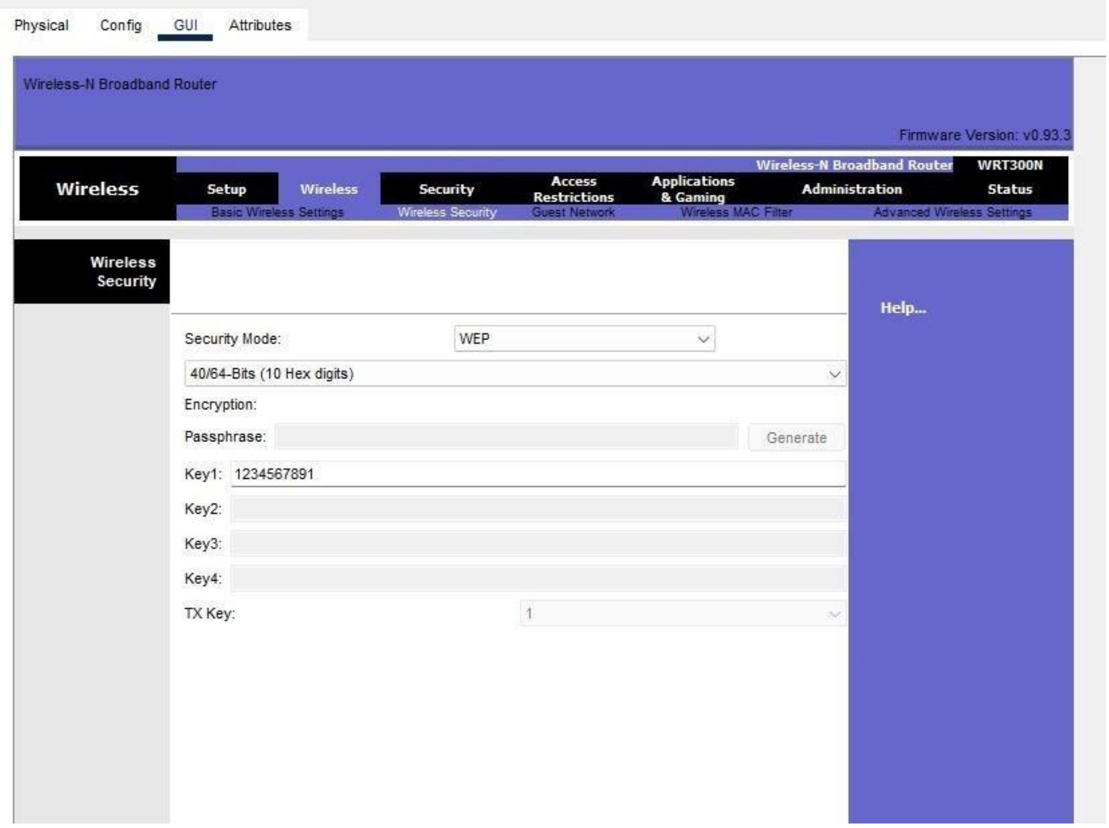


Figure 86

There you will assign the WEP key on the wireless security.

WEP is deprecated due to vulnerabilities. Always use WPA2 or WPA3 in production networks. But the Cisco Packet Tracer, the modem that we are using only allows WEP security key.

Assignment #2:

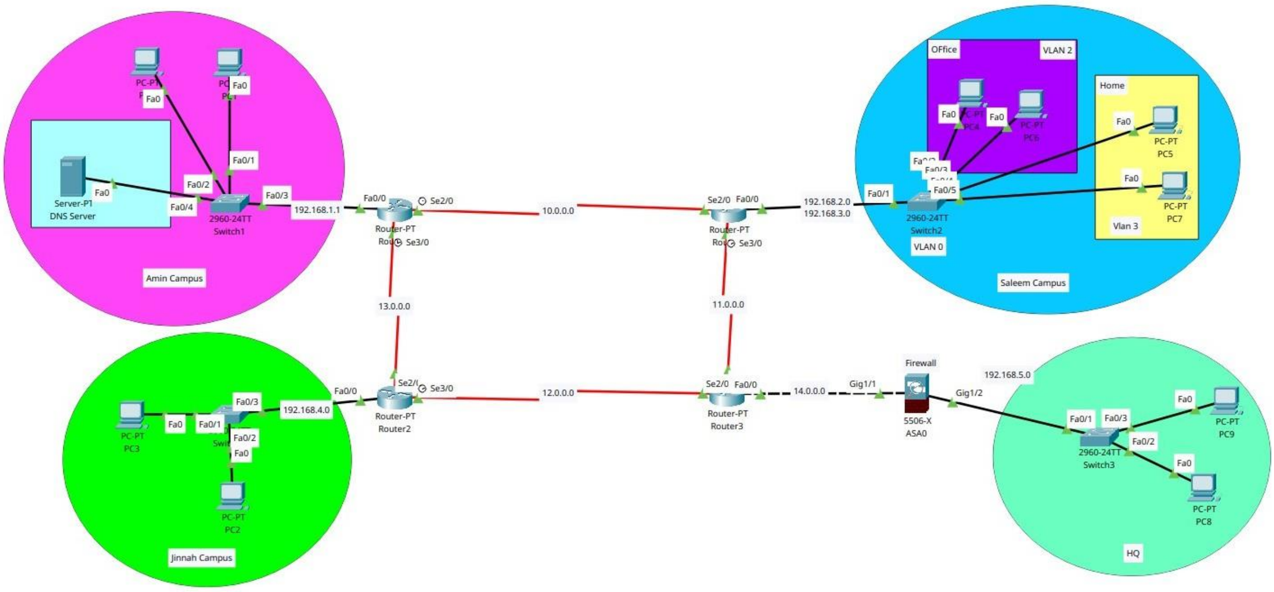


Figure 87

This network is a multi-campus enterprise setup with inter-campus communication facilitated via routers and secured through a firewall. The design uses VLANs, DHCP, DNS services, and dynamic routing to ensure efficient communication and segmentation. The topology includes four main campuses

Amin Campus:

- Server0 functioning as a DNS and DHCP server.
- PCs connected via Switch0.
- Connected to Router0 via FastEthernet.
- IP scheme: 192.168.1.0/24.
- Services: DHCP, DNS.

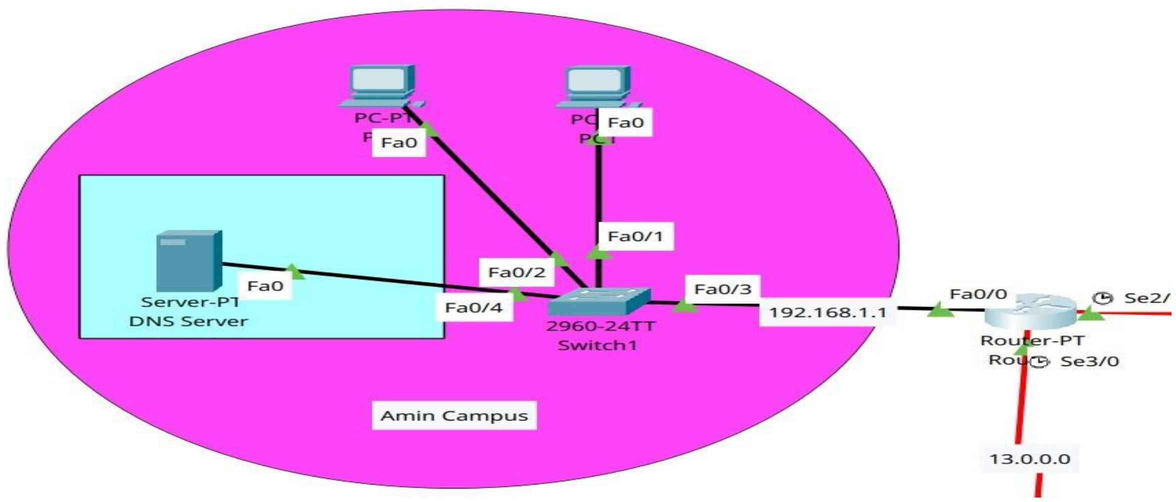


Figure 88

Where the server works as DHCP and DNS assigning the ip to the system also providing a DNS pathway to every system connected to the entire connection.

No.	Name	Type	Detail
0	mainpage	A Record	192.168.1.2

Figure 89

Ip POOL generated by the server

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
HQPool	192.168....	192.168....	192.168....	255.255....	253	0.0.0.0	0.0.0.0
AminPOOL	192.168....	192.168....	192.168....	255.255....	253	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168....	255.255....	512	0.0.0.0	0.0.0.0

Figure 90

Jinnah Campus

- Hosts several PCs connected through a Switch to Router2.
- Uses flat network.
- IP scheme: 192.168.4.0/24.

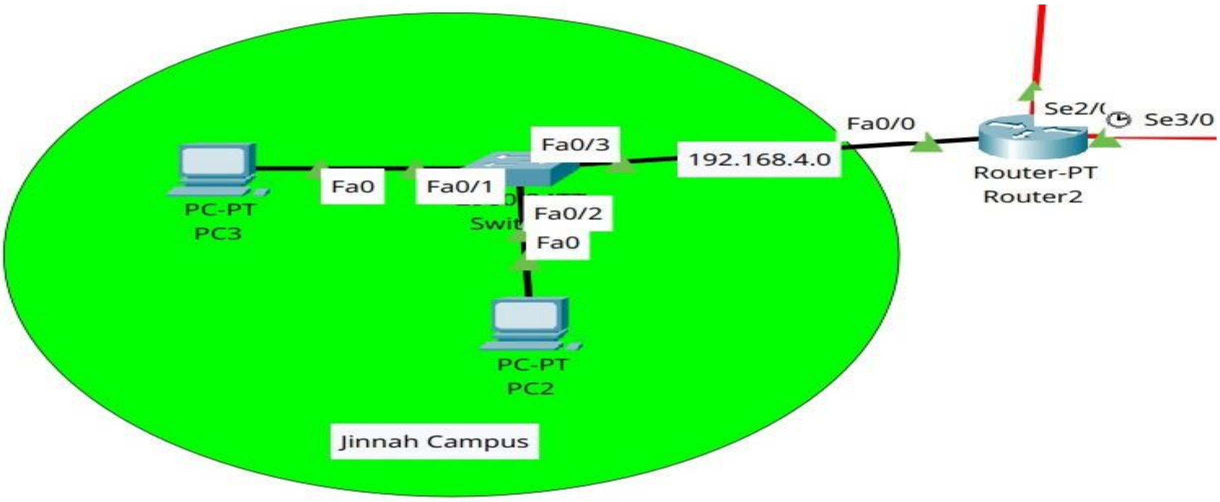


Figure 91

In the above router is responsible to generate the IP pool for this Network.

```
jinnah>show ip dhcp pool

Pool jinnahpool :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 0
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.4.1          192.168.4.1      - 192.168.4.254     2    / 0     / 254
jinnah>
```

Figure 92

Salem Campus

- Segmented into multiple VLANs: VLAN 2 (Office) — Includes PC0, PC1. VLAN 3 (Home) — Includes PC2, PC3.
- Connected through Switch1 to Router3 (Saleemc).
- Inter-VLAN routing is configured using sub-interfaces on Router3.
- Uses DHCP pools for each VLAN.
- IP ranges:
- VLAN 2: 192.168.2.0/24
- VLAN 3: 192.168.3.0/24
- Default DNS: 192.168.1.2.

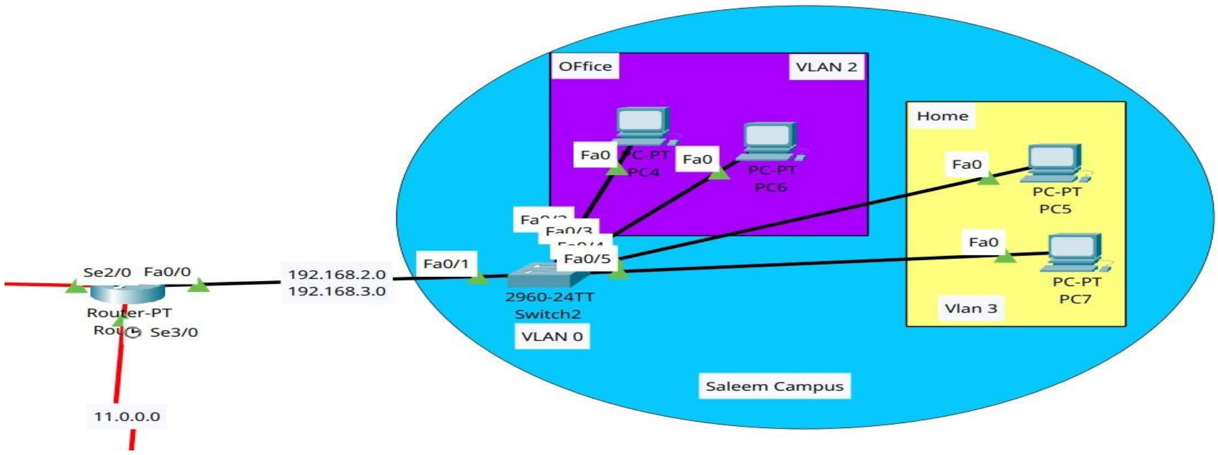


Figure 93

In this network the we have created a Virtual LANS that works on different configuration and perform there specific tasks.

The Switch is assigned the responsiblity on managing these VLANS. The Relevant Commands are given below.

```
Switch>en
Switch#config
Configuring from terminal, memory, or network [terminal]? t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name office
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name home
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#int range f0/2-3
Switch(config-if-range)#siwtchport mode access
                        ^
% Invalid input detected at '^' marker.

Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#int range f0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 3
Switch(config-if-range)#int f0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#wr
                 ^
% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#wr
```

Figure 94

where as router is responsible for managing the ip pool of the whole network.

```
saleem>show ip dhcp pool

Pool VLAN2_pool :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 2
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.2.1          192.168.2.1      - 192.168.2.254     2    / 2     / 254

Pool VLAN3_POOL :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Excluded addresses             : 2
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.3.1          192.168.3.1      - 192.168.3.254     0    / 2     / 254
saleem>
```

Figure 95

HQ Firewall Campus:

- Cisco ASA firewall is placed between Router1 and HQ.
- Interfaces:
- Gig1/1 (Inside) connected to Saleem Campus. Gig1/0 (Outside) connected to Router1.
- Provides traffic filtering and access control between internal and external networks.
- NAT and routing are likely applied here.

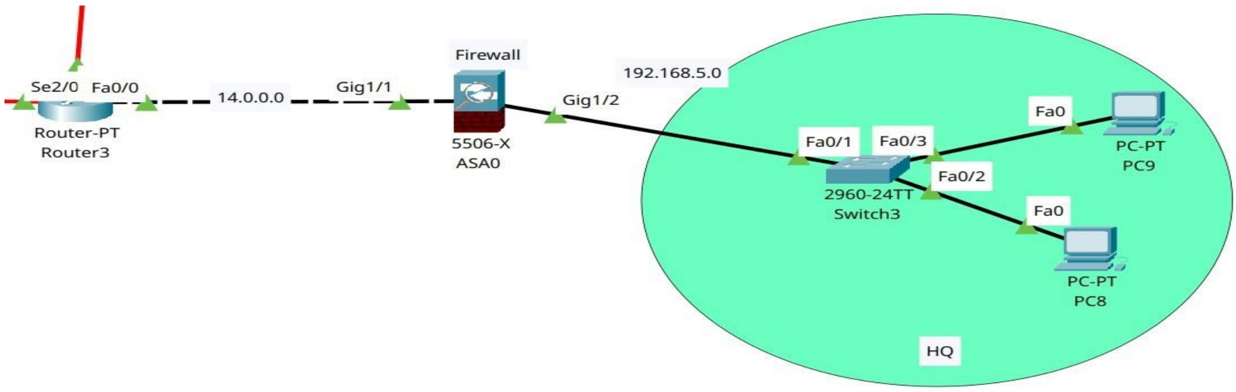


Figure 96

The firewall will act as a security barrier, filtering all incoming and outgoing traffic between external networks and the HQ Campus. It ensures that only authorized and safe data packets are allowed to pass through to the internal network. Additionally, the firewall is configured to function as a DHCP server, assigning IP addresses dynamically to the devices within the HQ Campus.

```
ciscoasa(config)#show running-config dhcpd
dhcpd dns 192.168.1.2
dhcpd option 3 ip 192.168.5.1
!
dhcpd address 192.168.5.1-192.168.5.10 inside
dhcpd enable inside
!
```

Figure 97

Main Connection with Routing:

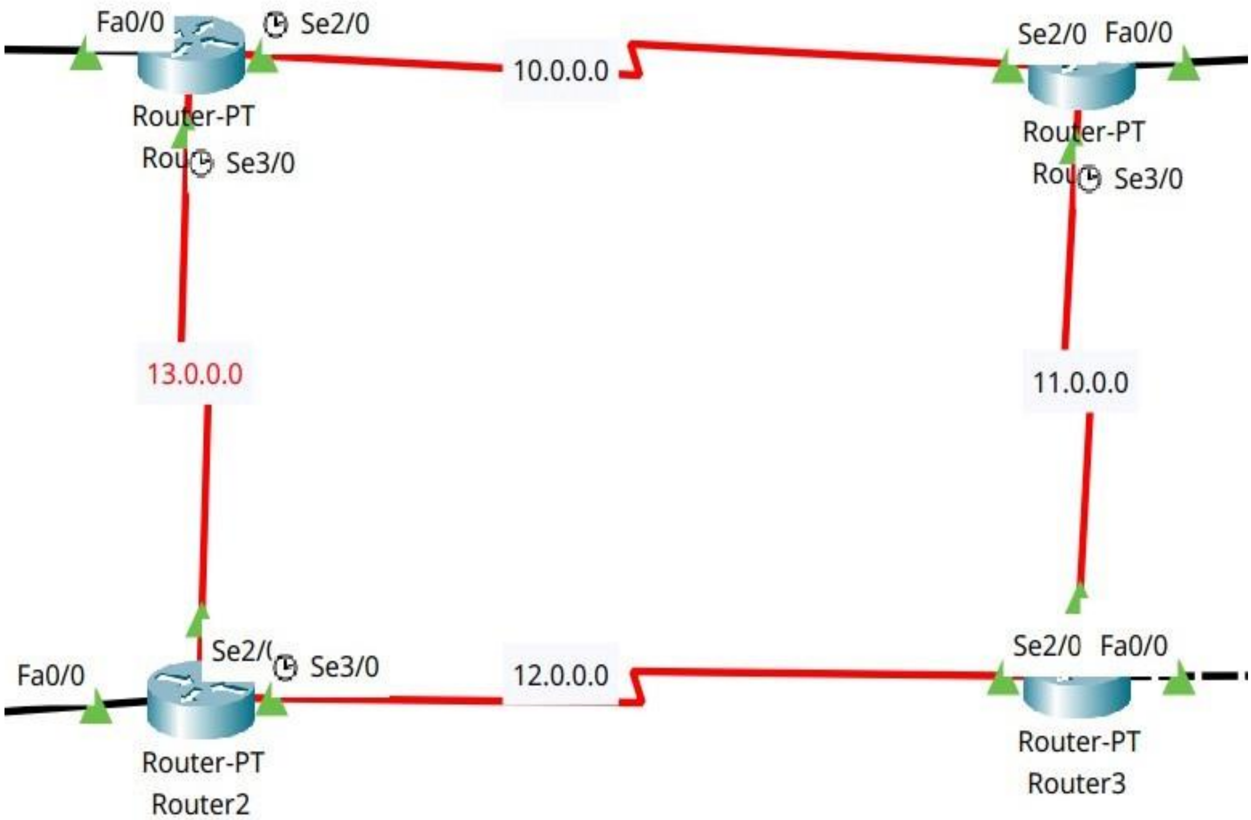


Figure 98

Routing

- All routers are using RIP v2 with no auto-summary.
- Routing networks include: 10.0.0.0/24
- 11.0.0.0/24
- 12.0.0.0/24
- 13.0.0.0/24
- Campus-specific LAN networks (192.168.x.0/24)

Amin Campus Router Configuration:

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname amin
amin(config)#int f0/0
amin(config-if)#ip add 192.168.1.1 255.255.255.0
```

```
amin(config-if)#no shutdown
amin(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up int
se2/0
amin(config-if)#ip add 10.0.0.1
% Incomplete command.
amin(config-if)#ip add 10.0.0.1 255.255.255.0
amin(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial2/0, changed state to down
amin(config-if)#int se3/0
amin(config-if)#ip add 13.0.0.1 255.255.255.0
amin(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial3/0, changed state to down
amin(config-if)#router rip
amin(config-router)#version 2
amin(config-router)#no auto-summary
amin(config-router)#network 192.168.1.0
amin(config-router)#network 10.0.0.0
amin(config-router)#network 13.0.0.0
amin(config-router)#do wr
Building configuration... [OK]
amin(config-router)#exit
amin(config)#
```

Saleem Campus Configuration:

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname saleem
saleem(config)#int f0/0
saleem(config-if)#no shut
saleem(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
saleem(config-if)#int se2/0
saleem(config-if)#ip add 10.0.0.2 255.255.255.0
saleem(config-if)#no shut
saleem(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
saleem(config-if)#int se3/0
saleem(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
saleem(config-if)#ip add 11.0.0.1 255.255.255.0
saleem(config-if)#no shut
%LINK-5-CHANGED: Interface Serial3/0, changed state to down
saleem(config-if)#router rip
```

```
saleem(config-router)#version 2
saleem(config-router)#no auto-summary
saleem(config-router)#network 192.168.2.0
saleem(config-router)#network 10.0.0.0
saleem(config-router)#network 11.0.0.0
saleem(config-router)#int f0/0.2
saleem(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up
saleem(config-subif)#encapsulation dot1Q 2
saleem(config-subif)#ip add 192.168.2.1 255.255.255.0
saleem(config-subif)#int f0/0.3
saleem(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up
saleem(config-subif)#encapsulation dot1Q 3
saleem(config-subif)#ip add 192.168.3.1 255.255.255.0
saleem(config-subif)#exit
saleem(config)#ip dhcp pool VLAN2_pool
saleem(dhcp-config)#network 192.168.2.0 255.255.255.0
saleem(dhcp-config)#default-router 192.168.2.1
saleem(dhcp-config)#dns-server 192.168.1.2
saleem(dhcp-config)#ip dhcp pool VLAN3_POOL
saleem(dhcp-config)#network 192.168.3.0 255.255.255.0
saleem(dhcp-config)#default-router 192.168.3.1
saleem(dhcp-config)#dns-server 192.168.1.2
saleem(dhcp-config)#ip dhcp excluded-address 192.168.2.1
saleem(config)#ip dhcp excluded-address 192.168.3.1
saleem(config)#exit
saleem#
%SYS-5-CONFIG_I: Configured from console by console saleem#wr
```

Switch Side of Saleem Campus Network Configuration:

```
Switch>en
Switch#config
Configuring from terminal, memory, or network [terminal]? t Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name office
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name home
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#int range f0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 2
```

```
Switch(config-if-range)#int range f0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 3
Switch(config-if-range)#int f0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Switch(config-if)#exit
%SYS-5-CONFIG_I: Configured from console by console wr
Building configuration... [OK]
```

Jinnah Campus Configuration:

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname jinnah
jinnah(config)#int f0/0
jinnah(config-if)#ip add 192.168.4.1 255.255.255.0
jinnah(config-if)#no shutdown
jinnah(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
jinnah(config-if)#int se3/0
jinnah(config-if)#ip add 13.0.0.2 255.255.255.0
jinnah(config-if)#no shut
%LINK-5-CHANGED: Interface Serial3/0, changed state to down jinnah(config-if)#int se2/0
jinnah(config-if)#ip add 12.0.0.2 255.255.255.0
jinnah(config-if)#no shut
jinnah(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
jinnah(config-if)#router rip
jinnah(config-router)#version 2
jinnah(config-router)#no auto-summary
jinnah(config-router)#network 192.168.4.0
jinnah(config-router)#network 13.0.0.0
jinnah(config-router)#network 12.0.0.0
jinnah(config-router)#exit
jinnah(config)#ip dhcp pool jinnahpool
jinnah(dhcp-config)#network 192.168.4.0 255.255.255.0
jinnah(dhcp-config)#default-router 192.168.4.1
jinnah(dhcp-config)#dns-server 192.168.1.2
jinnah(dhcp-config)#exit   jinnah(config)#exit
jinnah#wr
%SYS-5-CONFIG_I: Configured from console by console wr
Building configuration... [OK]
```

HQ Configuration:

```
Router>en Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname headoffice
headoffice(config)#int f0/0
headoffice(config-if)#ip add 14.0.0.1 255.255.255.0
headoffice(config-if)#no shut
headoffice(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
headoffice(config-if)#int  se2/0
headoffice(config-if)#ip add 12.0.0.1 255.255.255.0
headoffice(config-if)#no shut
headoffice(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
headoffice(config-if)#int se3/0
headoffice(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
headoffice(config-if)#ip add 11.0.0.2 255.255.255.0 headoffice(config-if)#no shut
headoffice(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
headoffice(config-if)#router rip
headoffice(config-router)#v
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up ersion 2
headoffice(config-router)#no auto-summary
headoffice(config-router)#network 14.0.0.0
headoffice(config-router)#network 12.0.0.0
headoffice(config-router)#network 13.0.0.0
headoffice(config-router)#network 11.0.0.0
headoffice(config-router)#exit
headoffice(config)#do wr
Building configuration...
```

Firewall Configuration:

```
ciscoasa>en Password:
ciscoasa#conf t
ciscoasa(config)#int g1/1
ciscoasa(config-if)#nameif outside
INFO: Security level for "outside" set to 0 by default. ciscoasa(config-if)#ip add 14.0.0.2
255.255.255.0
ciscoasa(config-if)#no shut
ciscoasa(config-if)#int gig1/2
ciscoasa(config-if)#nameif inside
INFO: Security level for "inside" set to 100 by default.
ciscoasa(config-if)#ip add 192.168.5.1 255.255.255.0
ciscoasa(config-if)#no shutdown
ciscoasa(config-if)#route outside 0.0.0.0 0.0.0.0 14.0.0.1
ciscoasa(config)#object network INSIDE-NET
ciscoasa(config-network-object)#subnet 192.168.4.0 255.255.255.0 ciscoasa(config-network-
object)#nat (inside, outside) dynamic interface ciscoasa(config-network-object)#
ciscoasa(config-network-object)#
ciscoasa(config-network-object)#access-list OUTSIDE-IN extended permit icmp any any
ciscoasa(config)#access-list OUTSIDE-IN extended permit udp any any eq 53
ciscoasa(config)#access-group OUTSIDE-IN in interface outside ciscoasa(config)#dhcpd ?
configure mode commands/options:
address      Configure the IP pool address range after this keyword auto_config Enable auto
configuration from client
dns          Configure the IP addresses of the DNS servers after this keyword domain
             Configure DNS domain name after this keyword
enable       Enable the DHCP server
lease        Configure the DHCPD lease length after this keyword
option       Configure options to pass to DHCP clients after this keyword
ciscoasa(config)#dhcpd address 192.168.5.1 ?
configure mode commands/options:
outside      Name of interface GigabitEthernet1/1 inside      Name of interface GigabitEthernet1/2
ciscoasa(config)#dhcpd address 192.168.5.1-192.168.5.10 inside
ciscoasa(config)#dhcpd option 3 ip 192.168.5.1
ciscoasa(config)#dhcpd dns 192.168.1.2
ciscoasa(config)#dhcpd enable inside
```

Lab 12

Experiment Name: Multiplayer Switches

Aim: To design and configure a VLAN-based network topology using a Layer 2 switch and a multilayer switch for inter-VLAN routing, enabling communication between different subnets and centralized server access.

Apparatus (Software): Cisco Packet Tracer.

Description:

In this experiment, a network topology is created using Cisco Packet Tracer that demonstrates the implementation of VLANs to segment network traffic between different groups of PCs. Two separate VLANs are configured on a Layer 2 switch to logically divide the network. A multilayer switch is used to perform inter-VLAN routing, allowing communication between the VLANs. A centralized server is connected to the multilayer switch to provide shared services across the network. This setup enhances network security, improves performance, and illustrates the functionality of VLANs and Layer 3 switching in modern network designs.

Figure 99

We are using this switch because it has a better compatibilty with VLAN configuration on layer 3. with the help of this switch we can easily route and perform the task of VLAN configuration and DNS configuration.

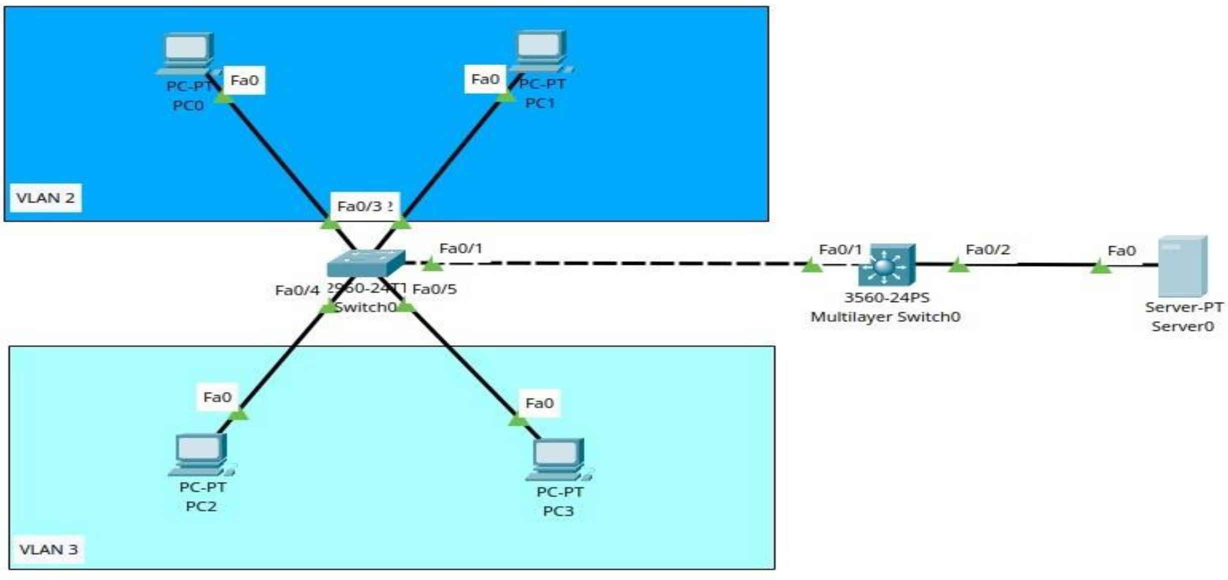


Figure 100

Having two switches of different architecture means that different level of configuration is required for both. lets start with the configuration of 2 Layer Switch.
First open the CLI of the switch.

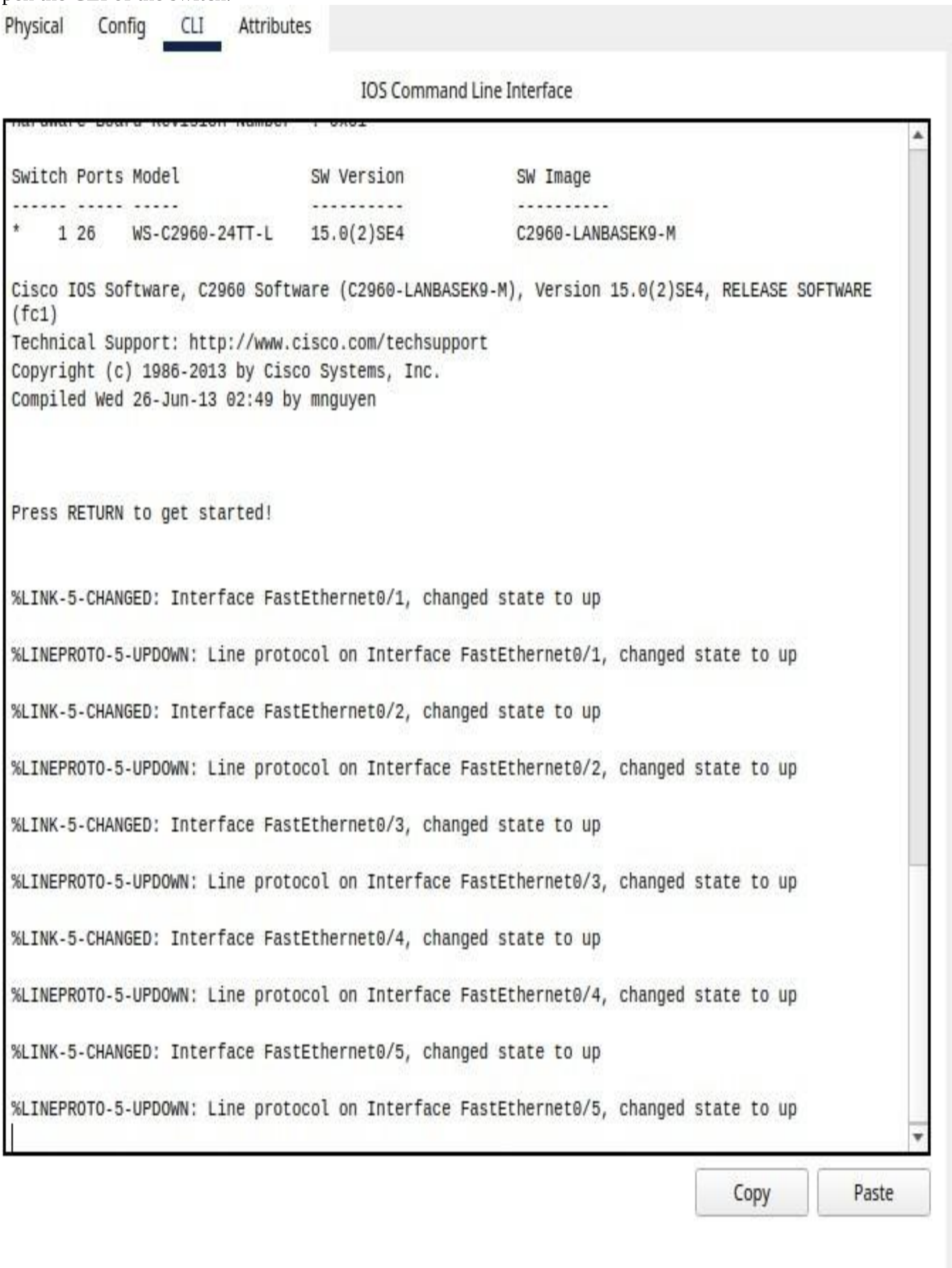


Figure 101

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Layer2Switch
Layer2Switch(config)#VLAN 10
Layer2Switch(config-vlan)#name IT
Layer2Switch(config-vlan)#exit
Layer2Switch(config)#VLAN 20
Layer2Switch(config-vlan)#name HR
Layer2Switch(config-vlan)#exit
Layer2Switch(config)#int range f0/2-3
Layer2Switch(config-if-range)#switchport access VLAN 10
Layer2Switch(config-if-range)#int range f0/4-5
Layer2Switch(config-if-range)#switchport access VLAN 20
Layer2Switch(config-if-range)#int f0/1
Layer2Switch(config-if)#switchport mode trunk

Layer2Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Layer2Switch(config-if)#exit
Layer2Switch(config)#exit
Layer2Switch#
%SYS-5-CONFIG_I: Configured from console by console

Layer2Switch#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   IT                               active    Fa0/2, Fa0/3
20   HR                               active    Fa0/4, Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Layer2Switch#wr
Building configuration...
[OK]
```

Figure 102

after creating all the vlans and assigning them ports its time to configure the multilayer switch. you can assign IP pool here but our task require us to get the ip pool directly from the dns server.

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname multiswitch
multiswitch(config)#vlan 10
multiswitch(config-vlan)#name it
multiswitch(config-vlan)#vlan 20
multiswitch(config-vlan)#name HR
multiswitch(config-vlan)#vlan 10
multiswitch(config-vlan)#name IT
multiswitch(config-vlan)#exit
multiswitch(config)#int f0/1
multiswitch(config-if)#switchport trunk encapsulation dot1Q
multiswitch(config-if)#switchport mode trunk
multiswitch(config-if)#exit
multiswitch(config)#int vlan 10
multiswitch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

multiswitch(config-if)#ip add
multiswitch(config-if)#ip address 192.168.10.1 255.255.255.0
multiswitch(config-if)#no shutdown
multiswitch(config-if)#int vlan 20
multiswitch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

multiswitch(config-if)#ip add 192.168.20.1 255.255.255.0
multiswitch(config-if)#no shut
multiswitch(config-if)#exit
multiswitch(config)#int vlan 1
multiswitch(config-if)#ip add 192.168.1.1 255.255.255.0
multiswitch(config-if)#no shut

multiswitch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

multiswitch(config-if)#exit
multiswitch(config)#ip routing
multiswitch(config)#int vlan 1
multiswitch(config-if)#ip helper-address 192.168.1.1
multiswitch(config-if)#int vlan 10
multiswitch(config-if)#ip helper-address 192.168.1.2
multiswitch(config-if)#int vlan 20
multiswitch(config-if)#ip helper-address 192.168.1.3
multiswitch(config-if)#exit
multiswitch(config)#do wr
Building configuration...
[OK]
```

Figure 103

after setting up the multilayer switch we will head up to the server where we will assign the DNS and we will setup DHCP services.

lets first assign the IP to the DNS server.

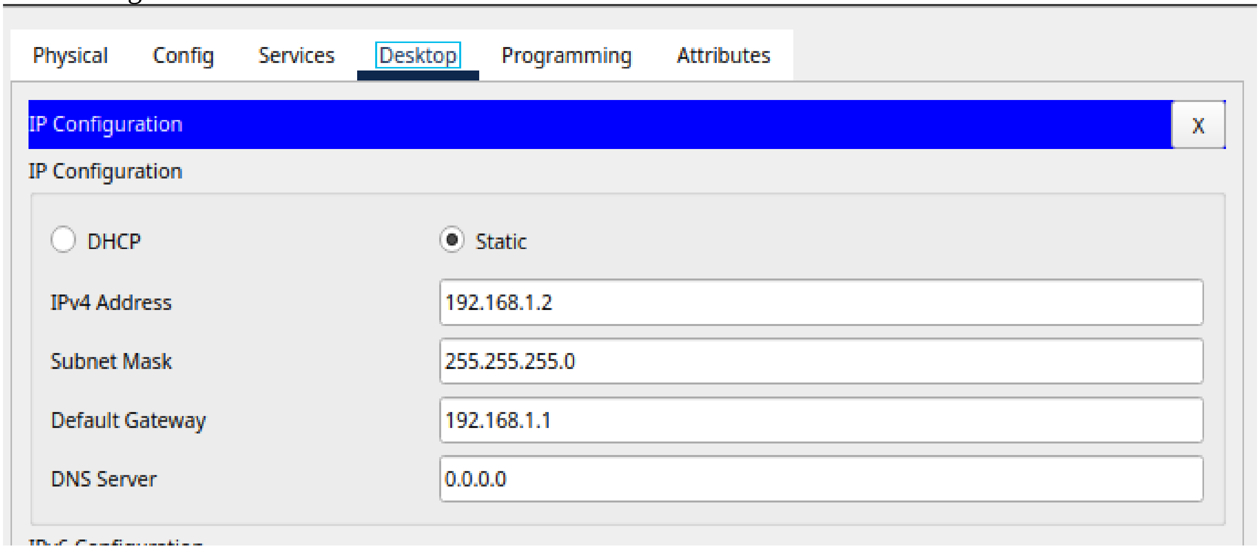


Figure 104

then go to services tab and navigate to the DNS services there u will enable the services.

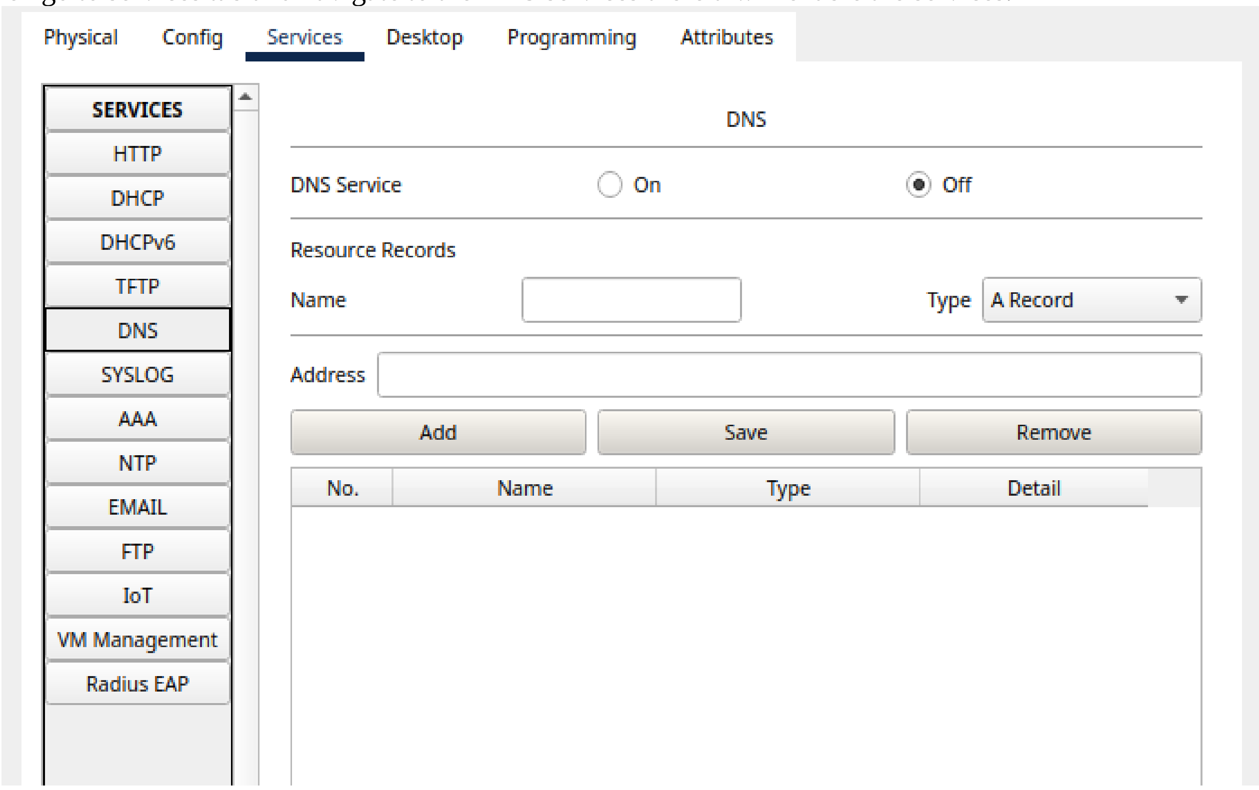


Figure 105

Then you will enter the name and address of the server and then you will hit add that way you will enable the dns service and you will create a path to access that DNS.
it will look like this

No.	Name	Type	Detail
0	main page	A Record	192.168.1.2

Figure 106

after setting up the DNS service lets set up the DHCP service as well. lets first navigate to the DHCP tab. and lets enable them.

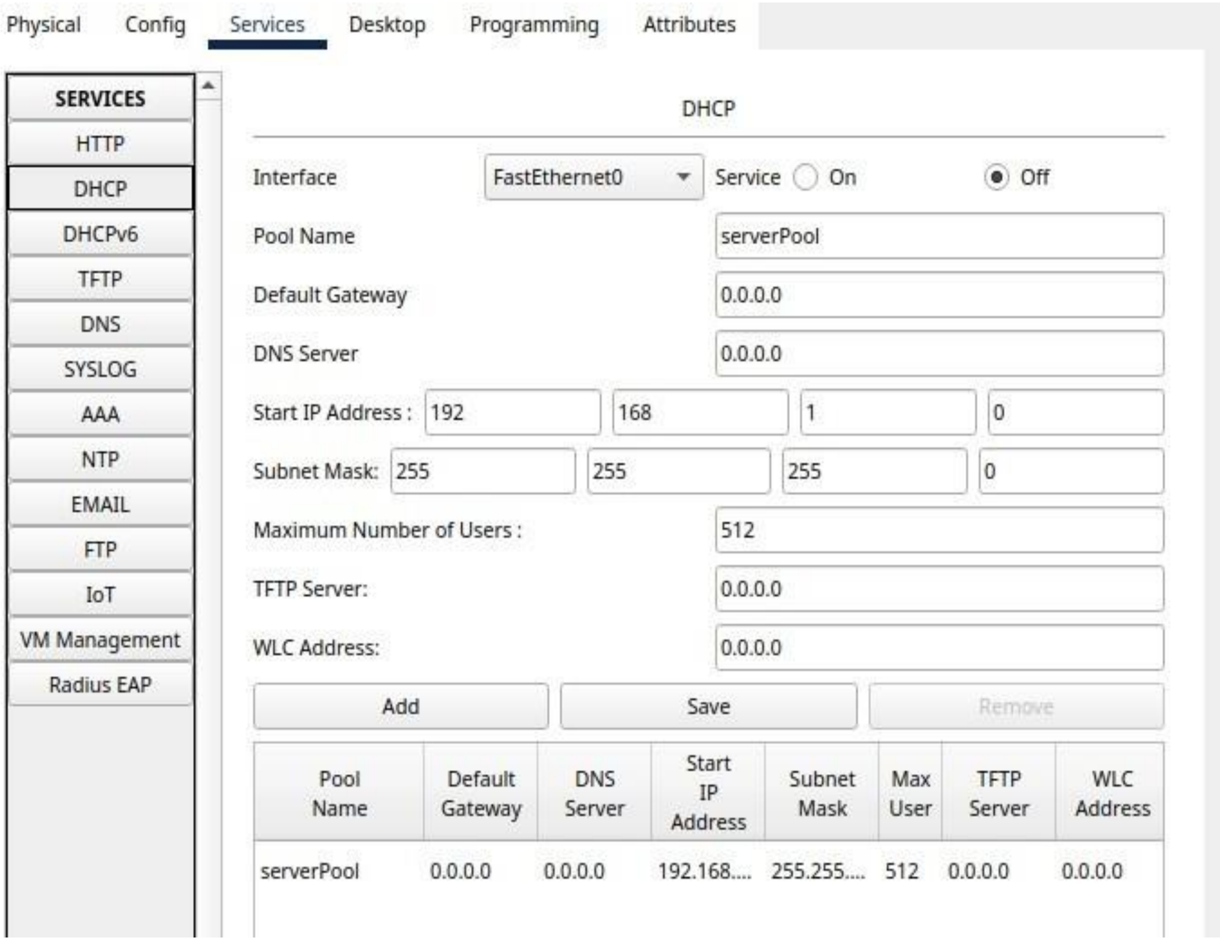


Figure 107

after enabling change the pool name to what we want it to be. and then change the gateway to the main gateway that you want (it would be different for each VLANS).

the dns would be the server ip and then you will tell the server from where it will start the allocation of the ip. then you will hit the add and you will create the pool.

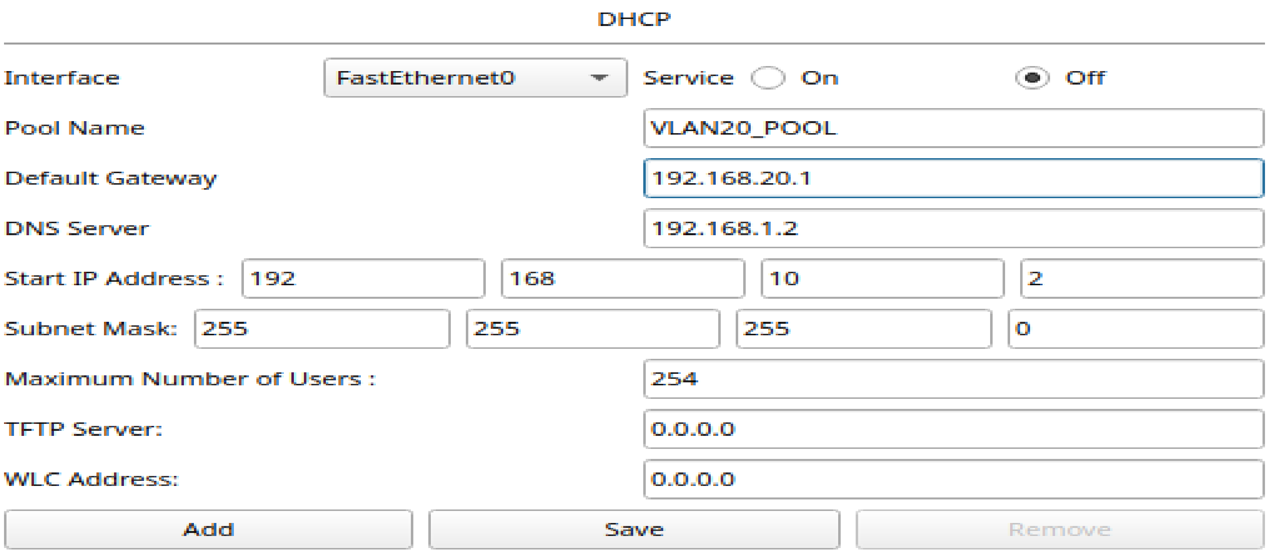


Figure 108

the whole pool would look like this

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
VLAN20_POOL	192.168.20.1	192.168.1.2	192.168.20.2	255.255.255.0	254	0.0.0.0	0.0.0.0
VLAN10_POOL	192.168.10.1	192.168.1.2	192.168.10.2	255.255.255.0	254	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.1.0	255.255.255.0	512	0.0.0.0	0.0.0.0

Figure 109

after doing this lets head to the system and see whether we can assign the pool through dhcp or not.

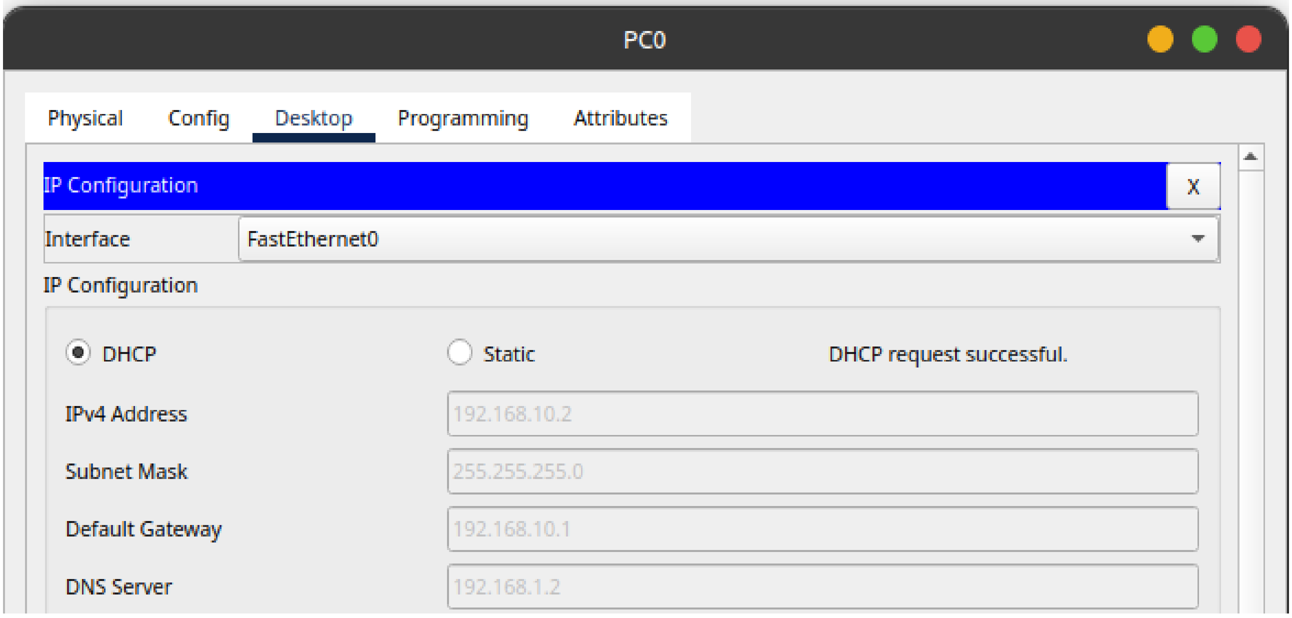


Figure 110

we were able to achieve it.
lets check whether the DNS works or not.

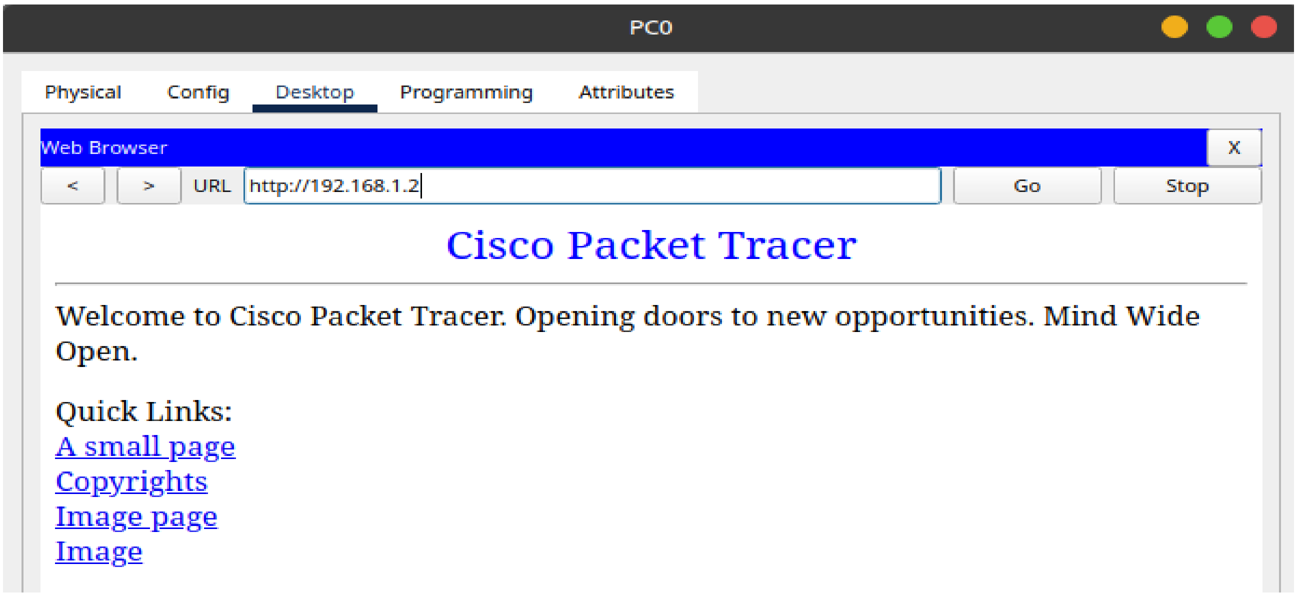


Figure 111

Lab 13

Experiment Name: Dynamic Routing

Aim: To design and configure a multi-router network topology that enables communication between different subnets using OSPF and RIP for dynamic routing.

Apparatus (Software): Cisco Packet Tracer.

Description:

To facilitate efficient routing and communication between different subnets, Open Shortest Path First (OSPF), Rip, dynamic routing protocosl, is implemented across all routers. OSPF and RIP allows the routers to automatically learn and update routes, ensuring optimal path selection and adaptability to network changes. Each router is configured with appropriate interface IP addresses, and OSPF routing is set up to advertise connected networks. The final configuration enables end-to-end connectivity across all PCs in the topology, demonstrating the practical use of dynamic routing in a statically addressed multi-router environment.

Diagram:

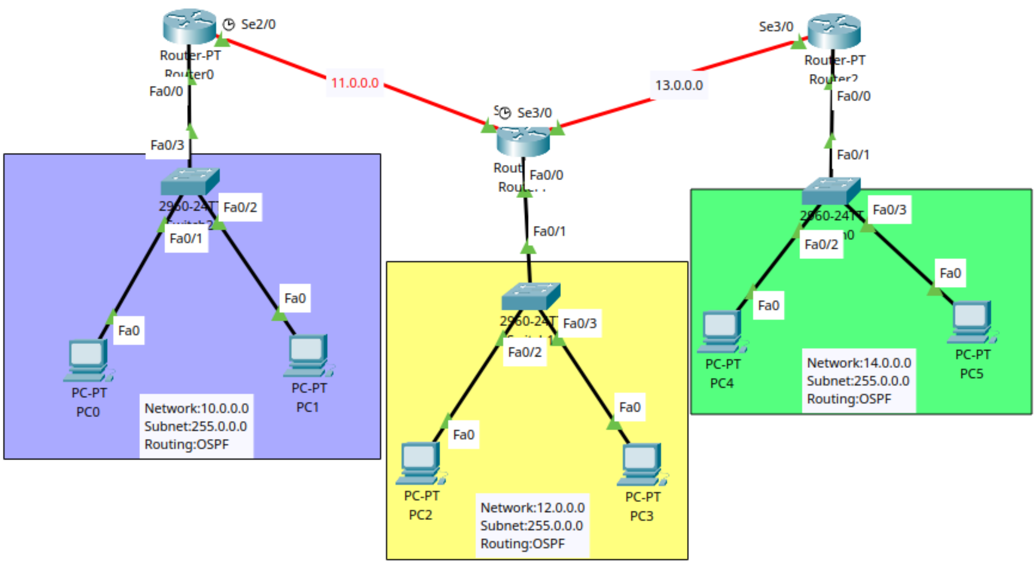


Figure 112

OSPF Installation:

Open Shortest Path First (OSPF) is a dynamic routing protocol used within large or complex networks to automatically discover and maintain routes between routers. It is a link-state protocol that calculates the shortest path to each network using Dijkstra's algorithm. OSPF supports hierarchical routing, allows for fast convergence, and is widely used in enterprise networks.

In configuration, routers are assigned to an OSPF process ID and networks are advertised using the network command under OSPF router configuration mode. All routers within the same OSPF area (commonly Area 0) share routing information and build a consistent view of the network topology.

Router0:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
```

```
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#ip add 10.0.0.1 255.0.0.0
Router(config-if)#int s2/0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#ip add 11.0.0.1 255.0.0.0
Router(config-if)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config)#router OSPF 1
Router(config-router)#network 10.0.0.0 0.255.255.255
% Incomplete command.
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#network 11.0.0.0 0.255.255.255 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
00:08:22: %OSPF-5-ADJCHG: Process 1, Nbr 13.0.0.1 on Serial2/0 from LOADING to FULL, Loading Done
```

Router 1:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip add 12.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int s2/0
Router(config-if)#ip add 11.0.0.2 255.0.0.0
Router(config-if)#no shut
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config-if)#int s3/0
Router(config-if)#ip add 13.0.0.1 255.0.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#network 11.0.0.0 0.255.255.255 area 0
Router(config-router)#network 12.0.0.0 0.255.255.255 area 0
00:08:21: %OSPF-5-ADJCHG: Process 1, Nbr 11.0.0.1 on Serial2/0 from LOADING to FULL,
Loading Done

Router(config-router)#network 13.0.0.0 0.255.255.255 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
00:09:08: %OSPF-5-ADJCHG: Process 1, Nbr 14.0.0.1 on Serial3/0 from LOADING to FULL, Loading
Done
```

Router 3:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int s3/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

Router(config-if)#ip add 13.0.0.2 255.0.0.0
Router(config-if)#int f0/
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip add 14.0.0.1 255.0.0.0
Router(config-if)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 14.0.0.0 0.255.255.255 area 0
Router(config-router)#network 13.0.0.0 0.255.255.255 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
00:09:07: %OSPF-5-ADJCHG: Process 1, Nbr 13.0.0.1 on Serial3/0 from LOADING to FULL, Loading Done
```

RIP Installation:

Routing Information Protocol (RIP) is one of the earliest and simplest dynamic routing protocols used in IP networks. It is a distance-vector protocol that uses hop count as its routing metric, where each hop between routers counts as one. RIP supports a maximum of 15 hops, making it suitable for smaller networks.
RIP routers periodically exchange their full routing tables with neighbors every 30 seconds, helping the network dynamically adapt to changes. RIP version 2 (RIPv2), the modern version, supports classless routing (CIDR), multicast updates, and better security.
RIP is easy to configure and ideal for learning and small-scale deployments where simplicity is prioritized over speed or scalability.

Router 0:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#ip add 10.0.0.1 255.0.0.0
Router(config-if)#int s2/0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
```

```
Router(config-if)#ip add 11.0.0.1 255.0.0.0
Router(config-if)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 10.0.0.0
Router(config-router)# network 11.0.0.0
Router(config-router)# no auto-summary
Router(config-router)# exit
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
00:08:22: %OSPF-5-ADJCHG: Process 1, Nbr 13.0.0.1 on Serial2/0 from LOADING to FULL, Loading Done
```

Router 1:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip add 12.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int s2/0
Router(config-if)#ip add 11.0.0.2 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config-if)#int s3/0
Router(config-if)#ip add 13.0.0.1 255.0.0.0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 11.0.0.0
Router(config-router)# network 12.0.0.0
Router(config-router)# network 13.0.0.0
Router(config-router)# no auto-summary
Router(config-router)# exit
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
00:09:08: %OSPF-5-ADJCHG: Process 1, Nbr 14.0.0.1 on Serial3/0 from LOADING to FULL, Loading
Done
```

Router 2:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int s3/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

Router(config-if)#ip add 13.0.0.2 255.0.0.0
Router(config-if)#int f0/
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip add 14.0.0.1 255.0.0.0
Router(config-if)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 13.0.0.0
Router(config-router)# network 14.0.0.0
Router(config-router)# no auto-summary
```

Router(config-router)# exit
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
00:09:07: %OSPF-5-ADJCHG: Process 1, Nbr 13.0.0.1 on Serial3/0 from LOADING to FULL, Loading Done

Total Packets Shared in the Topology:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC4	Router2	ICMP		0.000	N	0	(edit)	
	Successful	PC4	Router1	ICMP		0.000	N	1	(edit)	
	Successful	PC4	Router0	ICMP		0.000	N	2	(edit)	
	Successful	PC2	Router2	ICMP		0.000	N	3	(edit)	
	Successful	PC2	PC4	ICMP		0.000	N	4	(edit)	

Figure 113

Ping Also Working on the Packet Tracer:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 14.0.0.2

Pinging 14.0.0.2 with 32 bytes of data:

Reply from 14.0.0.2: bytes=32 time=18ms TTL=126
Reply from 14.0.0.2: bytes=32 time=1ms TTL=126
Reply from 14.0.0.2: bytes=32 time=1ms TTL=126
Reply from 14.0.0.2: bytes=32 time=22ms TTL=126

Ping statistics for 14.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 22ms, Average = 10ms

C:\>
```

Figure 114